T.B.C. : CDS-MATH-ALL-26-A Serial No. **940280**	**TEST BOOKLET** **[A]** **COMPLETE REPOSITORY**	**Elementary Mathematics** **Marks: +1.000 / -0.333**

UNION PUBLIC SERVICE COMMISSION

COMBINED DEFENCE SERVICES (CDS) EXAMINATION

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EXAMINATION SESSION & PAPER TITLE	ITEMS	SERIES
UPSC CDS 2 2026 (CDS II) : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
UPSC CDS 1 2026 (CDS I) : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (II) 2025 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (I) 2025 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (II) 2024 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (I) 2024 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (II) 2023 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]
CDS (I) 2023 : Elementary Mathematics Real Test Paper	**100 Qs**	Series [A]

SECTION : UPSC CDS 2 2026 (CDS II) : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

1. If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (II) 2026 Q1]
(a) 24 (b) —25 (c) 28 (d) —

2. If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (II) 2026 Q2]
(a) 60 (b) 50 (c) 45 (d)

3. For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (II) 2026 Q3]
(a) , B and C (b) , B and D
(c) , E and F 41. The number of three digit numbers (d) Six

4. What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (II) 2026 Q4]
(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E
(c) , C and D only (d) and E

5. What is the mean of the values ? [CDS (II) 2026 Q5]
(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

6. The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [CDS (II) 2026 Q6]
(a) nM — k (b) n (c) 81 (d) 105

7. What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [CDS (II) 2026 Q7]
(a) 81 (b) 105 (c) 144 (d) 243

8. If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (II) 2026 Q8]
(a) 500 (b) 300 (c) 60 (d) 40

9. If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (II) 2026 Q9]
(a) 50 (b) 55 (c) 60 (d) 65 15 A -

10. What is the arithmetic mean ? [CDS (II) 2026 Q10]
(a) 25 (b) 50 (c) 75 (d) 25

11. Which one of the following is a factor of the [CDS (II) 2026 Q11]
(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

12. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (II) 2026 Q12]
(a) 54 days (b) 60 days
(c) 64 days (d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5

13. A tin of oil was — full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was — fill]. Oil of. how many EX)ttles can the tin contain ? (All bottles are of equal volurne) [CDS (II) 2026 Q13]
(a) 35 (b) 40 (c) 45 (d) 50

14. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (II) 2026 Q14]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

15. If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [CDS (II) 2026 Q15]
(a) 24 (b) —25 (c) 28 (d) —

16. What is the median Of the following data ? [CDS (II) 2026 Q16]
(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E
(c) , C and D only (d) and E

17. What is the median of the distribu- tion ? [CDS (II) 2026 Q17]
(a) 9 (b) 5 (c) 9 (d) 3

18. What is the mode of the distribution ? [CDS (II) 2026 Q18]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

19. The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (II) 2026 Q19]
(a) 25 (b) 50 (c) 75 (d) 25

20. If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (II) 2026 Q20]
(a) 20 (b) 25
(c) 40 (d) 45 Consider the following types of data

21. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (II) 2026 Q21]
(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and

— SPACE FOR ROUGH WORK —

22. What is the mean of the ages of minor children ? [CDS (II) 2026 Q22]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

23. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (II) 2026 Q23]

(a) only (b) 2 only

(c) oth I and (d) Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15, (b) 11 (c) 12 (d)

24. X is as given below : x frequency [CDS (II) 2026 Q24]

(a) 36 (b) 44 (c) 52 64. (d) 150

25. The frequency distribution of a variate X is as given below : [CDS (II) 2026 Q25]

(a) 36 (b) 44 (c) 52 64. (d) 150

26. If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (II) 2026 Q26]

(a) 45 (b) 48 (c) 51 (d) 55

27. Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (II) 2026 Q27]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

28. If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (II) 2026 Q28]

(a) 24 (b) —25 (c) 28 (d) —

29. If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (II) 2026 Q29]

(a) 60 (b) 50 (c) 45 (d)

30. For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (II) 2026 Q30]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

31. What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (II) 2026 Q31]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

32. What is the mean of the values ? [CDS (II) 2026 Q32]

(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

33. The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [CDS (II) 2026 Q33]

(a) nM — k (b) n (c) 81 (d) 105

34. What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [CDS (II) 2026 Q34]

(a) 81 (b) 105 (c) 144 (d) 243

35. If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (II) 2026 Q35]

(a) 500 (b) 300 (c) 60 (d) 40

36. A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? [CDS (II) 2026 Q36]

(a) 2 (b) 225 (c) 2 (d) 2

37. If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (II) 2026 Q37]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

38. What is the arithmetic mean ? [CDS (II) 2026 Q38]

(a) 25 (b) 50 (c) 75 (d) 25

39. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (II) 2026 Q39]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

40. The sum of the areas of AAOD and ABOC is equal to the sum of the areas of MOB and ADOC [CDS (II) 2026 Q40]

(a) 924 square units (b) 824 square units

(c) 624 square units (d) 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

41. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (II) 2026 Q41]

(a) 54 days (b) 60 days

(c) 64 days (d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5

— SPACE FOR ROUGH WORK —

42. The quantity which must be added to (1— x 2) to obtain x 3 is 2X3 + 3X2 + I 2x3+x + x —1 (b) Consider the following statements • [CDS (II) 2026 Q42]

(a) 15 (b) 25 (c) 20 (d) 30 9. 10.

43. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (II) 2026 Q43]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

44. If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [CDS (II) 2026 Q44]

(a) 24 (b) —25 (c) 28 (d) —

45. What is the median Of the following data ? [CDS (II) 2026 Q45]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

46. What is the median of the distribu- tion ? [CDS (II) 2026 Q46]

(a) 9 (b) 5 (c) 9 (d) 3

47. What is the mode of the distribution ? [CDS (II) 2026 Q47]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

48. The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (II) 2026 Q48]

(a) 25 (b) 50 (c) 75 (d) 25

49. If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (II) 2026 Q49]

(a) 20 (b) 25

(c) 40 (d) 45 Consider the following types of data

50. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (II) 2026 Q50]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and

51. What is the mean of the ages of minor children ? [CDS (II) 2026 Q51]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

52. Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [CDS (II) 2026 Q52]

(a) umulative distribution in more than type (b) umulåtive distribution in less than type

(c) ontinuous frequency distri- bu tion (d) None Of above / 26A

53. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (II) 2026 Q53]

(a) only (b) 2 only

(d) Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15,

(c) oth I and (b) 11 (c) 12 (d)

54. X is as given below : x frequency [CDS (II) 2026 Q54]

(a) 36 (b) 44 (c) 52 64. (d) 150

55. The frequency distribution of a variate X is as given below : [CDS (II) 2026 Q55]

(a) 36 (b) 44 (c) 52 64. (d) 150

56. If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (II) 2026 Q56]

(a) 45 (b) 48 (c) 51 (d) 55

57. Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (II) 2026 Q57]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

58. If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (II) 2026 Q58]

(a) 24 (b) —25 (c) 28 (d) —

59. If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (II) 2026 Q59]

(a) 60 (b) 50 (c) 45 (d)

60. For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (II) 2026 Q60]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

61. What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (II) 2026 Q61]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

— SPACE FOR ROUGH WORK —

62. The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [CDS (II) 2026 Q62]

(a) nM — k (b) n (c) 81 (d) 105

63. What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [CDS (II) 2026 Q63]

(a) 81 (b) 105 (c) 144 (d) 243

64. If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (II) 2026 Q64]

(a) 500 (b) 300 (c) 60 (d) 40

65. A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? [CDS (II) 2026 Q65]

(a) 2 (b) 225 (c) 2 (d) 2

66. If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (II) 2026 Q66]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

67. What is the arithmetic mean ? [CDS (II) 2026 Q67]

(a) 25 (b) 50 (c) 75 (d) 25

68. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (II) 2026 Q68]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

69. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (II) 2026 Q69]

(a) 54 days (b) 60 days

(c) 64 days (d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5

70. Which of the following is/are correctly matched ? [CDS (II) 2026 Q70]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

71. If A+B= equal to ? x2 —4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the [CDS (II) 2026 Q71]

(a) 30 (b) 45 (c) 60 (d) 75

72. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (II) 2026 Q72]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

73. If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [CDS (II) 2026 Q73]

(a) 24 (b) —25 (c) 28 (d) —

74. What is the median Of the following data ? [CDS (II) 2026 Q74]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

75. What is the median of the distribu- tion ? [CDS (II) 2026 Q75]

(a) 9 (b) 5 (c) 9 (d) 3

76. What is the mode of the distribution ? [CDS (II) 2026 Q76]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

77. The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (II) 2026 Q77]

(a) 25 (b) 50 (c) 75 (d) 25

78. If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (II) 2026 Q78]

(a) 20 (b) 25

(c) 40 (d) 45 Consider the following types of data

79. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (II) 2026 Q79]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and

80. What is the mean of the ages of minor children ? [CDS (II) 2026 Q80]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

81. Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [CDS (II) 2026 Q81]

(a) umulative distribution in more than type (b) umulåtive distribution in less than type

(c) ontinuous frequency distri- bu tion (d) None Of above / 26A

— SPACE FOR ROUGH WORK —

82. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (II) 2026 Q82]

(a) only (b) 2 only

(c) oth I and (d) Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15, (b) 11 (c) 12 (d)

83. X is as given below : x frequency [CDS (II) 2026 Q83]

(a) 36 (b) 44 (c) 52 64. (d) 150

84. The frequency distribution of a variate X is as given below : [CDS (II) 2026 Q84]

(a) 36 (b) 44 (c) 52 64. (d) 150

85. If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (II) 2026 Q85]

(a) 45 (b) 48 (c) 51 (d) 55

86. Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (II) 2026 Q86]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

87. If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (II) 2026 Q87]

(a) 24 (b) —25 (c) 28 (d) —

88. The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to (d) 385 7 [CDS (II) 2026 Q88]

(a) 24 (b) —25 (c) 28 (d) —

89. If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (II) 2026 Q89]

(a) 60 (b) 50 (c) 45 (d)

90. For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (II) 2026 Q90]

(a) , B and C (b) , B and D

(c) , E and F 41. The number of three digit numbers (d) Six

91. What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (II) 2026 Q91]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

92. What is the mean of the values ? [CDS (II) 2026 Q92]

(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

93. The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [CDS (II) 2026 Q93]

(a) nM — k (b) n (c) 81 (d) 105

94. What is the geometric mean of 3, 9, 27, 81, 243, 729, 2187? [CDS (II) 2026 Q94]

(a) 81 (b) 105 (c) 144 (d) 243

95. If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (II) 2026 Q95]

(a) 500 (b) 300 (c) 60 (d) 40

96. A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? [CDS (II) 2026 Q96]

(a) 2 (b) 225 (c) 2 (d) 2

97. If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (II) 2026 Q97]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

98. What is the arithmetic mean ? [CDS (II) 2026 Q98]

(a) 25 (b) 50 (c) 75 (d) 25

99. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (II) 2026 Q99]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

100. a, — + a, —a, + as is divisible by [CDS (II) 2026 Q100]

(a) I only (b) 2 only (c) oth 1 and (d) Neither I nor 2 What is x

SECTION : UPSC CDS 1 2026 (CDS I) : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

101. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (I) 2026 Q1]

(a) 5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? (b) 4 square units

(c) 245 square units (d) r square units

— SPACE FOR ROUGH WORK —

102. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —l [CDS (I) 2026 Q2]
(a) 242 cm (b) 385 cm (c) 462 cm (d) 770 cm

103. cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (I) 2026 Q3]
(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

104. A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (I) 2026 Q4]
(a) 63•5 cm (b) 65 cm (c) 5 cm (d) 75 cm

105. Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (I) 2026 Q5]
(a) 60 square units (b) 62 square units
(c) 64 square units (d) nnot be determined due to insufficient data

106. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (I) 2026 Q6]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

107. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (I) 2026 Q7]
(a) 2280 (b) 2640 (c) 3820 (d) 4130

108. Consider the following statements with reference to the given figure : [CDS (I) 2026 Q8]
(a) 924 square units (b) 824 square units
(c) 624 square units (d) 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

109. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (I) 2026 Q9]
(a) 100/3 (b) 40 (c) 50 (d) 200/3

110. If then what•is one of the values of x equal to? [CDS (I) 2026 Q10]
(a) 15 (b) 25 (c) 20 (d) 30 9. 10.

111. A person saves more than he did the previous year. If he saves *2000 in the first year, in how many years will he save 170000 ? [CDS (I) 2026 Q11]
(a) 16 years (b) 17 years (c) 18 years (d) 19 years

112. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (I) 2026 Q12]
(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

113. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (I) 2026 Q13]
(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th
(c) 15m (d) l •5m

114. 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (I) 2026 Q14]
(a) 12 (b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/ (d) 14xa

115. Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (I) 2026 Q15]
(a) only (b) only (c) oth A and B (d) neither A nor B

116. What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (I) 2026 Q16]
(a) The triangle is isosceles (b) The triangle is equilateral
(c) None Of the ab ove For the next three (d) 3124

117. The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (I) 2026 Q17]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

118. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [CDS (I) 2026 Q18]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

119. Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (I) 2026 Q19]
(a) 4445 cm (b) 16 cm (c) 4745 cm (d) 476 cm

120. If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (I) 2026 Q20]
(a) 539 cm3 (b) 616 cm3 (c) 1078 cm3 (d) 1232 cm3

121. What is the slant height of the cone ? [CDS (I) 2026 Q21]
(a) 60 m (b) 65 m (c) 68 m (d) 70m

122. A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (I) 2026 Q22]
(a) 40m cm' (b) 4500 cm3 (c) 5000 cm3 (d) m3

123. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (I) 2026 Q23]
(a) 5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ? (b) 4 square units
(c) 245 square units (d) r square units

— SPACE FOR ROUGH WORK —

124. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —l [CDS (I) 2026 Q24]
(a) 242 cm (b) 385 cm (c) 462 cm (d) 770 cm

125. cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (I) 2026 Q25]
(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

126. A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (I) 2026 Q26]
(a) 63•5 cm (b) 65 cm (c) 5 cm (d) 75 cm

127. Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (I) 2026 Q27]
(a) 60 square units (b) 62 square units
(c) 64 square units (d) nnot be determined due to insufficient data

128. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (I) 2026 Q28]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

129. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (I) 2026 Q29]
(a) 2280 (b) 2640 (c) 3820 (d) 4130

130. The number 2x3*5x7x11+1 is [CDS (I) 2026 Q30]
(a) prime number (b) not a prime, but power of a prime
(c) not a power of a prime, but a composite even number (d) not a of a prime, but a composite odd number

131. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (I) 2026 Q31]
(a) 100/3 (b) 40 (c) 50 (d) 200/3

132. If 0.764 y 1.236 x, then what is the Y¯x value Of [CDS (I) 2026 Q32]
(a) x- (b) x =
(c) x = 11.5 If 0.764 y = 1.236 x, Y—x value Of (d) 0.764

133. In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? •l. The length Of the race is 300 m. [CDS (I) 2026 Q33]
(a) 2% (b) 8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs (d) Zero

134. If 96-64a3 3 = what is alt + 4a3 equal to ? then [CDS (I) 2026 Q34]
(a) 30 (b) 45 (c) 60 (d) 75

135. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (I) 2026 Q35]
(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

136. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (I) 2026 Q36]
(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th
(c) 15m (d) l •5m

137. 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (I) 2026 Q37]
(a) 12 (b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/ (d) 14xa

138. Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (I) 2026 Q38]
(a) only (b) only (c) oth A and B (d) neither A nor B

139. What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (I) 2026 Q39]
(a) The triangle is isosceles (b) The triangle is equilateral
(c) None Of the ab ove For the next three (d) 3124

140. The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (I) 2026 Q40]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

141. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [CDS (I) 2026 Q41]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

142. Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (I) 2026 Q42]
(a) 4445 cm (b) 16 cm (c) 4745 cm (d) 476 cm

143. If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (I) 2026 Q43]
(a) 539 cm3 (b) 616 cm3 (c) 1078 cm3 (d) 1232 cm3

144. What is the slant height of the cone ? [CDS (I) 2026 Q44]
(a) 60 m (b) 65 m (c) 68 m (d) 70m

145. A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (I) 2026 Q45]
(a) 40m cm' (b) 4500 cm3 (c) 5000 cm3 (d) m3

— SPACE FOR ROUGH WORK —

146. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (I) 2026 Q46]

(a) 5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?

(b) 4 square units

(c) 245 square units

(d) r square units

147. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —I [CDS (I) 2026 Q47]

(a) 242 cm (b) 385 cm (c) 462 cm (d) 770 cm

148. cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (I) 2026 Q48]

(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

149. A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (I) 2026 Q49]

(a) 63•5 cm (b) 65 cm (c) 5 cm (d) 75 cm

150. Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (I) 2026 Q50]

(a) 60 square units

(b) 62 square units

(c) 64 square units

(d) nnot be determined due to insufficient data

151. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (I) 2026 Q51]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

152. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (I) 2026 Q52]

(a) 2280 (b) 2640 (c) 3820 (d) 4130

153. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (I) 2026 Q53]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ?

(b) 9th

(c) 15m

(d) I •5m

154. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (I) 2026 Q54]

(a) 54 days

(b) 60 days

(c) 64 days

(d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5

155. D In the above figure, ABCD is a parallelo- gram. p is a point on BC such that PB : PC=I : [CDS (I) 2026 Q55]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

156. If BD is the perpendicular on the side AC, then what is the length of BD ? 1008 65 (b) cm 65 A - [CDS (I) 2026 Q56]

(a) I and 11 only

(b) 11 and Ill only

(c) I and Ill only

(d) 1, 11 and 111 Consider the following for the next two

157. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (I) 2026 Q57]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

158. If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [CDS (I) 2026 Q58]

(a) 24 (b) —25 (c) 28 (d) —

159. What is the median Of the following data ? [CDS (I) 2026 Q59]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?

(b) nd E

(c) , C and D only

(d) and E

160. What is the median of the distribu- tion ? [CDS (I) 2026 Q60]

(a) 9 (b) 5 (c) 9 (d) 3

161. What is the mode of the distribution ? [CDS (I) 2026 Q61]

(a) I and 11 only

(b) 11 and Ill only

(c) I and Ill only

(d) 1, 11 and 111 Consider the following for the next two

162. The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (I) 2026 Q62]

(a) 25 (b) 50 (c) 75 (d) 25

163. for 160 men ? [CDS (I) 2026 Q63]

(a) 54 days (b) 60 days (c) 64 days (d) 72 days

164. If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (I) 2026 Q64]

(a) 20

(b) 25

(c) 40

(d) 45 Consider the following types of data

165. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (I) 2026 Q65]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and

166. What is the mean of the ages of minor children ? [CDS (I) 2026 Q66]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

— SPACE FOR ROUGH WORK —

167. Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [CDS (I) 2026 Q67]

(a) umulative distribution in more than type
(b) umulåtive distribution in less than type
(c) ontinuous frequency distri- bu tion
(d) None Of above / 26A

168. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (I) 2026 Q68]

(a) only
(b) 2 only
(c) oth I and
(d) Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15, (b) 11 (c) 12 (d)

169. X is as given below : x frequency [CDS (I) 2026 Q69]

(a) 36 (b) 44 (c) 52 64. (d) 150

170. The frequency distribution of a variate X is as given below : [CDS (I) 2026 Q70]

(a) 36 (b) 44 (c) 52 64. (d) 150

171. If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (I) 2026 Q71]

(a) 45 (b) 48 (c) 51 (d) 55

172. Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (I) 2026 Q72]

(a) Mean Median
(b) Mean Mode
(c) Mean > Median
(d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

173. If each of n numbers x, — i (i = l, 2, 3, ...m) is replaced by (i + l)xi, then the new mean IS n(n + l) 2 (d) [CDS (I) 2026 Q73]

(a) 24 (b) —25 (c) 28 (d) —

174. The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to (d) 385 7 [CDS (I) 2026 Q74]

(a) 24 (b) —25 (c) 28 (d) —

175. If the mean age of combined group Of boys and girls is 18 years and-the mean Of age of boys is 20 and that of girls is 16; then what is the percentage of boys in the group ? [CDS (I) 2026 Q75]

(a) 60 (b) 50 (c) 45 (d)

176. For two observations, the sum is S and product is P. What is the harmonic mean Of these two observations ? (b) S A - [CDS (I) 2026 Q76]

(a) , B and C
(b) , B and D
(c) , E and F 41. The number of three digit numbers
(d) Six

177. What is the arithmetic mean of the first ten composite numbers? 102 (d) 112 [CDS (I) 2026 Q77]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?
(b) nd E
(c) , C and D only
(d) and E

178. What is the mean of the values ? [CDS (I) 2026 Q78]

(a) 38 (b) 40 (c) 42 (d) 44 GI. 62. 63. 64.

179. The arithmetic mean of n numbers is M. If the sum of first (n — 1) terms is k, then what is the nth number ? (b) nM — k (c) n(M — k) [CDS (I) 2026 Q79]

(a) nM — k (b) n (c) 81 (d) 105

180. If 5 is multiplied to each observation, then what Will be the new arithmetic mean? [CDS (I) 2026 Q80]

(a) 500 (b) 300 (c) 60 (d) 40

181. A distribution consists of 3 components With frequencies 45, 40 and 55 having their means 2, 2 • 5 and 2 respectively. What is the mean of the combined dis tribu tion? [CDS (I) 2026 Q81]

(a) 2 (b) 225 (c) 2 (d) 2

182. If 10 is added to each observation then what will be the new arithmetic mean ? [CDS (I) 2026 Q82]

(a) 50 (b) 55 (c) 60 (d) 65 15 A -

183. What is the arithmetic mean ? [CDS (I) 2026 Q83]

(a) 25 (b) 50 (c) 75 (d) 25

184. What is the weighted mean of first IO natural numbers whose weights are equal to the goffesponding numlYr ? (d) 4-5 [CDS (I) 2026 Q84]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I •5m

185. What is, the mean proportional between and (27 ? (b) 146 A stock Of food is enough for 240 men for 48 days. How long will the same stock last for 160 men ? [CDS (I) 2026 Q85]

(a) 54 days
(b) 60 days
(c) 64 days
(d) 72 days 9 2 3 = 2 and If x+Y x+y x —Y then what is ¯ equal to ? (a) 3/2 (c) 2/3 (d) 1/5

186. What is tlE median of the ages Of minor children ? (b) 5 years (c) 7 years (d) Cannot determined due to in- sufficient data [CDS (I) 2026 Q86]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

— SPACE FOR ROUGH WORK —

187. If a variable takes discrete values a + 4, a-3•5, a-2•5, a-3, a-2, a+O•5, a +5 and a —08 where a > O, then the median of the data set is [CDS (I) 2026 Q87]

(a) 24 (b) —25 (c) 28 (d) —

188. What is the median Of the following data ? [CDS (I) 2026 Q88]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

189. What is the median of the distribu- tion ? [CDS (I) 2026 Q89]

(a) 9 (b) 5 (c) 9 (d) 3

190. What is the mode of the distribution ? [CDS (I) 2026 Q90]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

191. The numbers 59, 53, 51, 43, 36, (8x+ l), (x2+ l), 13, 12, 9, 7, 3 are written in descending order and their median is [CDS (I) 2026 Q91]

(a) 25 (b) 50 (c) 75 (d) 25

192. If each number is first multiplied by 2 and then increased by 5, then what is the mean Of new numbers ? [CDS (I) 2026 Q92]

(a) 20 (b) 25

(c) 40 (d) 45 Consider the following types of data

193. Marks of students who appeared for a test of 100 marks. Collar sizes of200 shirts sold in a week. Monthly incomes of250 employees Of a factory. For which of the above data, mode is a suitable measure of central tendency ? [CDS (I) 2026 Q93]

(a) I and 2 only (b) 2 Only (c) I and 3 only (d) 1 , 2 and

194. What is the mean of the ages of minor children ? [CDS (I) 2026 Q94]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

195. Consider •the following frequency distributién • Class 0-10 0-20 0-30 0-40 0-50 Frequency 3 What is the above frequency distributioÖ known as? [CDS (I) 2026 Q95]

(a) umulative distribution in more than type (b) umulåtive distribution in less than type

(c) ontinuous frequency distri- bu tion (d) None Of above / 26A

196. Histogram is an equivalent graphical representation of the frequency distribution. Histogram is suitable for continuous random variables, where the total frequency Of an interval is evenly distributed over the interval. Which . of the statements given above is/ are correct? [CDS (I) 2026 Q96]

(a) only (b) 2 only

(d) Neither I nor 2 What is the median Of the values • 11, 7, 6, 9, 12, 15,

(c) oth I and (b) 11 (c) 12 (d)

197. X is as given below : x frequency [CDS (I) 2026 Q97]

(a) 36 (b) 44 (c) 52 64. (d) 150

198. The frequency distribution of a variate X is as given below : [CDS (I) 2026 Q98]

(a) 36 (b) 44 (c) 52 64. (d) 150

199. If 15 is added to each of the first forty values and 5 is subtracted from each of the rernaining sixty values, the new mean becomes : [CDS (I) 2026 Q99]

(a) 45 (b) 48 (c) 51 (d) 55

200. Consider the following distribution Value of the variable 1 2 3 4 5 naaaa Frequency For What value Of f, is the arithmetic mean of the above distribution 3.1 ? [CDS (I) 2026 Q100]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

SECTION : CDS (II) 2025 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

201. x y, 17 : 19 I LCM 1615 t, (x + y) [CDS (II) 2025 Q1]

(a) Option A (b) Option B

(c) Option C (d) 36 180 360 — -E-k,åP r s t?

202. If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (II) 2025 Q2]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

203. The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (II) 2025 Q3]

(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

204. ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (II) 2025 Q4]

(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

205. Later on 5 grace marks are added to each student. What are the average marks of the revised marks Of the students? 30 31 32 1 P.T.O. [CDS (II) 2025 Q5]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

— SPACE FOR ROUGH WORK —

206. If a equation xa — x — 1 O, then what is equal to? [CDS (II) 2025 Q6]
(a) 56 (b) 60 cm (c) 64 cm (d) 1 only

207. I 15 30 I ? (b) 24 (c) 60 (d) 120 I. 100 1000 11. [CDS (II) 2025 Q7]
(a) Option A (b) 24 (c) 60 (d) 120 I. 100 1000 11.

208. What are these two categories ? [CDS (II) 2025 Q8]
(a) nd B (b) E and A (c) E and F (d) nd D

209. Consider a question and tqro statements : Question : Is always greater than Statement—I : x > y Statement—II : x 2 + > 2xy Which one Of the following is correct in respect Of the question and the State merits? [CDS (II) 2025 Q9]
(a) Statement—I alone is required to answer the question (b) Statement—II alone is required to answer the question
(c) oth Statemen and Statement—II are required to answer the question (d) Neither Statement—I nor State- ment—ll is required to answer the qu estion P.T.O.

210. Ifthe length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time ? [CDS (II) 2025 Q10]
(a) 150 (b) 300 (c) 450 (d) 600

211. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (II) 2025 Q11]
(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? (b) 47
(c) 43 (d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

212. ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (II) 2025 Q12]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

213. A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (II) 2025 Q13]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

214. 17. km/hr 2 km [CDS (II) 2025 Q14]
(a) km.mr (b) 2•75 knv'hr
(c) 3 kmJhr (d) 3•25 km/hr 18. 19. 20. xa + x2y — 10xy2 + 8y3

215. The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (II) 2025 Q15]
(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

216. ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (II) 2025 Q16]
(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

217. Which the Only 2 only Neither Q-DETA.N.FMN Stet-em correct ? [CDS (II) 2025 Q17]
(a) x = 1, y = (b) x = 2, y = (c) x = 3, y = (d) x = 1, y =

218. Which of the above statements is/are correct? [CDS (II) 2025 Q18]
(a) 1 only (b) 2 only
(c) oth 1 and (d) Neither nor 2 Which one is one of the factors of (a) (c) / 48A

219. Ihe present age of a father is equal to sum of the ages of his 4 children. Aner ten years the sum of the ages of the children will be 1-6 times the age of their father. What is the present age of father ? [CDS (II) 2025 Q19]
(a) 36 years (b) 40 years (c) 42 years (d) 45 years

220. Between 3 and 4 0'clock, both hour hand and minute hand will coincide past 3 0'clock between [CDS (II) 2025 Q20]
(a) 16 minutes (b) 17 minutes (c) 18 minutes (d) 19 minutes

221. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (II) 2025 Q21]
(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? (b) 47
(c) 43 (d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

222. ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (II) 2025 Q22]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

223. A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (II) 2025 Q23]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

224. If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (II) 2025 Q24]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

225. The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (II) 2025 Q25]
(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

226. ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (II) 2025 Q26]
(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

— SPACE FOR ROUGH WORK —

227. If p divi&s qr if p divides q, p must divi& r. 2' 2' WNch of the [CDS (II) 2025 Q27]

(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

228. If a set Acontains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does A u B contain ? [CDS (II) 2025 Q28]

(a) 130 (b) 100 (c) 80 (d) 70

229. If two factors of — 2a3 — 9a2 + 2a + 8 are (a + l) and (a — l), then what are the Other two factors ? [CDS (II) 2025 Q29]

(a) — 2) and (a + 4) (b) + 2) and (a + 4) (c) + 2) and (a — 4) (d) — 2) and (a —4)

230. How many zeros are there in the product 150 x 249 x 348* x 501 ? [CDS (II) 2025 Q30]

(a) 262 (b) 261 (c) 246 (d) 235

231. m, then the height Of the tower is close to [CDS (II) 2025 Q31]

(a) 86.6m (b) 84.6 m

(c) 82.6 m (d) 80.6 m 16 In a triangle ABC, LBCA = 600 and AB2 = BC2 + CA2 + X. What is X equal (a) (b) (c) (d) Zero

232. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (II) 2025 Q32]

(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? (b) 47

(c) 43 (d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

233. ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (II) 2025 Q33]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

234. A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (II) 2025 Q34]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

235. If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (II) 2025 Q35]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

236. The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (II) 2025 Q36]

(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

237. ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (II) 2025 Q37]

(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

238. If the average marks Of the complete class is 72, then what is the average marks Of the girls ? [CDS (II) 2025 Q38]

(a) 64 (b) 65 (c) 68 (d) 74

239. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (II) 2025 Q39]

(a) 3 years (b) 4 years (c) 5 years (d) 6 years

240. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (II) 2025 Q40]

(a) 2280 (b) 2640 (c) 3820 (d) 4130

241. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (II) 2025 Q41]

(a) 100/3 (b) 40 (c) 50 (d) 200/3

242. In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? •l. The length Of the race is 300 m. [CDS (II) 2025 Q42]

(a) 2% (b) 8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?

(c) 5<k$11 21. The number of pairs (d) Zero

243. The petpendiculars drawn from any point on OM to AB and AC are always equal Which of the above statements are correct ? [CDS (II) 2025 Q43]

(a) - 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) l, 2 and 3 The diagonal of a cube is 4 volume ? 2. and — + 3. 4. cm. What is its 5. 6.

244. Ift 2,600 is divided among three persons A, B and C in the ratio — : — , how much does A get ? [CDS (II) 2025 Q44]

(a) Option A (b) Option B

(c) Option C (d) 2 3 16 cm 32 cm 64 em 3 192 cm3 3 y 9 xy then 21 xy what where is the and Y value of x + y ?

245. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (II) 2025 Q45]

(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

246. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (II) 2025 Q46]

(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ? (b) 9th

(c) 15m (d) I •5m

— SPACE FOR ROUGH WORK —

247. 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (II) 2025 Q47]
(a) 12
(b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/
(d) 14xa

248. Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (II) 2025 Q48]
(a) only (b) only (c) oth A and B (d) neither A nor B

249. What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (II) 2025 Q49]
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) None Of the ab ove For the next three
(d) 3124

250. The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (II) 2025 Q50]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

251. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [CDS (II) 2025 Q51]
(a) will be 75. Let (b) 52% (c) 540/0 (d) 98 cm 108 cm2

252. Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (II) 2025 Q52]
(a) 4445 cm (b) 16 cm (c) 4745 cm (d) 476 cm

253. If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (II) 2025 Q53]
(a) 539 cm3 (b) 616 cm3 (c) 1078 cm3 (d) 1232 cm3

254. What is the slant height of the cone ? [CDS (II) 2025 Q54]
(a) 60 m (b) 65 m (c) 68 m (d) 70m

255. A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (II) 2025 Q55]
(a) 40m cm' (b) 4500 cm3 (c) 5000 cm3 (d) m3

256. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (II) 2025 Q56]
(a) 5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?
(b) 4 square units
(c) 245 square units
(d) r square units

257. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —l [CDS (II) 2025 Q57]
(a) 242 cm (b) 385 cm (c) 462 cm (d) 770 cm

258. cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (II) 2025 Q58]
(a) 343 cm2 (b) 350 cm2 (c) 686 cm2 (d) 700 cm2

259. A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (II) 2025 Q59]
(a) 63•5 cm (b) 65 cm (c) 5 cm (d) 75 cm

260. Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (II) 2025 Q60]
(a) 60 square units
(b) 62 square units
(c) 64 square units
(d) nnot be determined due to insufficient data

261. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (II) 2025 Q61]
(a) 3 years (b) 4 years (c) 5 years (d) 6 years

262. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (II) 2025 Q62]
(a) 2280 (b) 2640 (c) 3820 (d) 4130

263. What is Z BAP equal to ? [CDS (II) 2025 Q63]
(a) 300 (b) 400 (c) 450 (d) 600

264. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (II) 2025 Q64]
(a) 100/3 (b) 40 (c) 50 (d) 200/3

265. In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? •l. The length Of the race is 300 m. [CDS (II) 2025 Q65]
(a) 2%
(b) 8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs
(d) Zero

266. A person purchases one kg of tea powder from each of the four places A, B, C, D at the rate per 1 kg, 2 kg, 4 kg, 5 kg. If on an average he purchased x kg of tea powder per 1000, then what is the approximate value of x ? [CDS (II) 2025 Q66]
(a) 195 (b) 00 (c) 205 (d) 2•10

267. (d) 11 37. (d) [CDS (II) 2025 Q67]
(a) Option A
(b) Option B
(c) onsider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, IO and 12 cm : — 510 is divisible by 10 5 38. 39. 40. 12 What is the number of divisors of 360?
(d) 18

268. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (II) 2025 Q68]
(a) 300 cml (b) 350 cm2 (c) 375 cm2 (d) 400 cm2

— SPACE FOR ROUGH WORK —

269. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (II) 2025 Q69]
(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I •5m

270. 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (II) 2025 Q70]
(a) 12
(b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/
(d) 14xa

271. Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (II) 2025 Q71]
(a) only
(b) only
(c) oth A and B
(d) neither A nor B

272. What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (II) 2025 Q72]
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) None Of the ab ove For the next three
(d) 3124

273. The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (II) 2025 Q73]
(a) will be 75. Let
(b) 52%
(c) 540/0
(d) 98 cm 108 cm2

274. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [CDS (II) 2025 Q74]
(a) will be 75. Let
(b) 52%
(c) 540/0
(d) 98 cm 108 cm2

275. Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (II) 2025 Q75]
(a) 4445 cm
(b) 16 cm
(c) 4745 cm
(d) 476 cm

276. If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (II) 2025 Q76]
(a) 539 cm3
(b) 616 cm3
(c) 1078 cm3
(d) 1232 cm3

277. What is the slant height of the cone ? [CDS (II) 2025 Q77]
(a) 60 m
(b) 65 m
(c) 68 m
(d) 70m

278. A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (II) 2025 Q78]
(a) 40m cm'
(b) 4500 cm3
(c) 5000 cm3
(d) m3

279. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct ? (b) p 24 (c) 2p = q [CDS (II) 2025 Q79]
(a) 5 m 86. What is the area of the largest square plate cut from a circular disk Of radius one unit ?
(b) 4 square units
(c) 245 square units
(d) r square units

280. If the volume of the cylinder is 1617 CITI 3 , then what is the curved surface area of the cylinder? (Take It = —l [CDS (II) 2025 Q80]
(a) 242 cm
(b) 385 cm
(c) 462 cm
(d) 770 cm

281. cubes each of volume 343 cm3 are joined end to end. What is the total surface area of the resulting cuboid ? [CDS (II) 2025 Q81]
(a) 343 cm2
(b) 350 cm2
(c) 686 cm2
(d) 700 cm2

282. A tub is in the sharx of a frustum Of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height ? [CDS (II) 2025 Q82]
(a) 63•5 cm
(b) 65 cm
(c) 5 cm
(d) 75 cm

283. Let x, y, z (all prime) be the length, breadth and height respectively of a cuboid with x > y > z. The volume of the cuboid is 30k3 cubic units, where k is a natural number. What is the total surface area of the cuboid ? [CDS (II) 2025 Q83]
(a) 60 square units
(b) 62 square units
(c) 64 square units
(d) nnot be determined due to insufficient data

284. If u,.v, w are real numbers such that u3—8v3 18uvw, which one of the following is conect ? (d) u +2v=-3w [CDS (II) 2025 Q84]
(a) 3 years
(b) 4 years
(c) 5 years
(d) 6 years

285. W'hat is the difference between the sum of the cubes and that Of squares Of first ten natural numbers ? [CDS (II) 2025 Q85]
(a) 2280
(b) 2640
(c) 3820
(d) 4130

286. What are distinct prime mctors of the number 26381 ? [CDS (II) 2025 Q86]
(a) 29, 17,
(b) 31, 17,
(c) 19, 37,
(d) 23, 31,

287. If the height of a cone•is increased by 50%, then what is ihe percentage increase in the volunie Of the cone? [CDS (II) 2025 Q87]
(a) 100/3
(b) 40
(c) 50
(d) 200/3

288. If a Star figure is formed by elongating the sides of a regular pentagon, then the measure Of each angle at the angular points of the star figure is [CDS (II) 2025 Q88]
(a) 360
(b) 350
(c) 320
(d) 300

289. In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m. Which of the following is / are conect ? •l. The length Of the race is 300 m. [CDS (II) 2025 Q89]
(a) 2%
(b) 8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?
(c) 5<k$11 21. The number of pairs
(d) Zero

290. Three cubes each Of side 5 cm are joined end to end. What is the surface area Of the resulting cuboid ? [CDS (II) 2025 Q90]
(a) 300 cml
(b) 350 cm2
(c) 375 cm2
(d) 400 cm2

— SPACE FOR ROUGH WORK —

291. What is volume of the frustum of a cone with height h and radii r, , ? [CDS (II) 2025 Q91]
(a) 5 74. What is volume of the frustum of a cone with height h and radii r, , ?
(b) 9th
(c) 15m
(d) I •5m

292. 20 What is the difference in radii of the ends of the frustum ? (b) (d) None of the above What is the volume ofthe frustum ? [CDS (II) 2025 Q92]
(a) 12
(b) 13 If 196x4 = x6, then x3 is equal to which one of the following ?
(c) x'/
(d) 14xa

293. Let be a pyramid on a square base and B be a cube. Let a, b, c denote the number of edges, number of faces and number of corners respectively. Then the result a = b + c is true for [CDS (II) 2025 Q93]
(a) only
(b) only
(c) oth A and B
(d) neither A nor B

294. What is the surface area of the double cone so formed? (c) 1310-4 cm 2 (d) 13188 cm2 [CDS (II) 2025 Q94]
(a) The triangle is isosceles
(b) The triangle is equilateral
(c) None Of the ab ove For the next three
(d) 3124

295. The areas of three mutually perpendi- cular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to [CDS (II) 2025 Q95]
(a) will be 75. Let
(b) 52%
(c) 540/0
(d) 98 cm 108 cm2

296. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y-axis. 'Ihe cone is filled with water up to half of its height. The volume of water is 8 [CDS (II) 2025 Q96]
(a) will be 75. Let
(b) 52%
(c) 540/0
(d) 98 cm 108 cm2

297. Ifthesurface area ofa cube is 13254 cm2, then the length of its diagonal is [CDS (II) 2025 Q97]
(a) 4445 cm
(b) 16 cm
(c) 4745 cm
(d) 476 cm

298. If the total surface area is 616 cm2, what is the volume of the cylinder ? [CDS (II) 2025 Q98]
(a) 539 cm3
(b) 616 cm3
(c) 1078 cm3
(d) 1232 cm3

299. What is the slant height of the cone ? [CDS (II) 2025 Q99]
(a) 60 m
(b) 65 m
(c) 68 m
(d) 70m

300. A cinular a Of 20 cm its curved na is 1000 cml. What is ere volume Of [CDS (II) 2025 Q100]
(a) 40m cm'
(b) 4500 cm3
(c) 5000 cm3
(d) m3

SECTION : CDS (I) 2025 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

301. What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (I) 2025 Q1]
(a) — square cm
(b) 16vTä— — square cm
(c) (160 — 8r) square cm
(d) — square cm

302. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (I) 2025 Q2]
(a) 54 g
(b) 366 g
(c) 122 g
(d) 1087t g

303. The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) 2025 Q3]
(a) 15 cm
(b) 20 cm
(c) 25 cm
(d) 30 cm

304. A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) 2025 Q4]
(a) 22 m
(b) 20 m
(c) 18 m
(d) None Of the above

305. If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (I) 2025 Q5]
(a) It increases two times.
(b) It increases three times.
(c) It increases four times,
(d) It increases eight times.

306. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) 2025 Q6]
(a) 256
(b) 512
(c) 768
(d) 1024

307. p(x) q(x) (LCM) (HCF) —a3) (x2—ax+a2) I x4+a2x?+a4 t, q(x) [CDS (I) 2025 Q7]
(a) x4 —2a2x2 + a4
(b) x4 + 2a2x2+ a4
(c) x2—a2
(d) x4 —a2x2 + a4 x3+2x2-5X+2 x3 + 4x2 + 11. x+l

308. The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) 2025 Q8]
(a) litre
(b) litre ,
(c) — litre
(d) — litre

309. A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) 2025 Q9]
(a) 1540 cm3
(b) 1078cm3
(c) 1048 cm3
(d) 770 cm3

310. A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) 2025 Q10]
(a) 1330 cm2
(b) 1306 cm2
(c) 1296 cm2
(d) 1256 cm2

311. How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) 2025 Q11]
(a) 420
(b) 210
(c) 200
(d) 168 A - fiiü 12%

312. Q x, y t, (b) 24 (c) 27 (d) 36 15. 16. (b) 11 (c) 1 3fr(11ä 12 8 100m 20 1 16 10 gq, 200 m 60 cm [CDS (I) 2025 Q12]
(a) Option A
(b) 11
(c) 27
(d) 36 15. 16.

— SPACE FOR ROUGH WORK —

313. A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (I) 2025 Q13]

(a) — cm (b) — cm (c) — cm (d) — cm

314. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) 2025 Q14]

(a) 100 (b) 90 (c) 80 (d) 75

315. Consider the following statements : [CDS (I) 2025 Q15]

(a) (b) c equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) c (d) xyz

316. What is åe of bikers in mad 2017 ? [CDS (I) 2025 Q16]

(a) 163 (b) 152 (c) 141 (d) 134

317. 60 km/hr 48 km/hrGT (b) 12 #FTZ (c) 15 fiFTZ (d) 20 18. (downstream) x krn/hr@ (upstream) y km/hr I 3qq-Td ? x (c) (d) 19. 3TFT Q=x2+bx+c 11. x Q (b) 11 (d) 1, q fill 20. 21. 8 10 2727-940 - 379 ? [CDS (I) 2025 Q17]

(a) Option A (b) 12 #FTZ (c) Option C (d) 20 18.

318. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (I) 2025 Q18]

(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

319. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2025 Q19]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

320. In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) 2025 Q20]

(a) 2012 (b) 2013 (c) 2014 (d) 2015

321. If the annual income of X is 20% more than that of then the income of Y is less than that of X by pO/o. What is the value of p ? [CDS (I) 2025 Q21]

(a) 10 (b) 16i (c) 36 (d) 27

322. A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (I) 2025 Q22]

(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

323. Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (I) 2025 Q23]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

324. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) 2025 Q24]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

325. Colour of the eye Of the person [CDS (I) 2025 Q25]

(a) 21 (b) 24

(c) 36 (d) 42 Which one among the following is not correct ?

326. Number of times a }xrson correctly hits a target in ten attempts Which of the following in respect of the above is correct ? [CDS (I) 2025 Q26]

(a) I is a continuous variable, 2 is not a variable and 3 is a discrete variable (b) 1 is a continuous variable, 2 and 3 are discrete variables

(c) I and 3 are discrete variables and 2 is not a variable (d) •l, 2 and 3 all are discrete variables Consider the followiög •statements :

327. The data collected by investigator to be used by himself are called primary data [CDS (I) 2025 Q27]

(a) 21 (b) 24

(c) 36 (d) 42 Which one among the following is not correct ?

328. Consider the following : I. 10+ I [CDS (I) 2025 Q28]

(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and

329. Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (I) 2025 Q29]

(a) 264 (b) 201 (c) 132 (d) 105

330. Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (I) 2025 Q30]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

331. HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (I) 2025 Q31]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

332. What is the sum of the linear factors (in x and y) of the expression 2x-2 + — 3y2 ? [CDS (I) 2025 Q32]

(a) 2t-3y (b) 3x-2y (c) 3x + 2y (d) 2r+3y

333. For which values of k, does the equation x2 — kx + 2 0 have real and distinct solutions ? (b) only (c) k>2v5 only (d) k<-24i or k>2Vi [CDS (I) 2025 Q33]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

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334. If a + D + Y aß+ "y + ya, then what is (1 equal to? [CDS (I) 2025 Q34]

(a) only (b) k>2v5 only

(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

335. A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (I) 2025 Q35]

(a) 1010 (b) 1015 (c) 1020 (d) 1025

336. ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (I) 2025 Q36]

(a) 443 cm (b) 9 cm (c) I Ocm (d) m

337. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (I) 2025 Q37]

(a) 500 (b) 600 (c) 700 (d) 800

338. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2025 Q38]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

339. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (I) 2025 Q39]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

340. What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (I) 2025 Q40]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

341. The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (I) 2025 Q41]

(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

342. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (I) 2025 Q42]

(a) 120' (b) 135'

(c) 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is (d) following? aeroplane is Observed to be

343. What is the number of sides of the polygon? 9 (b) 10 (c) 11 (d) 12 ,'45A [CDS (I) 2025 Q43]

(a) 15 200 73. (b) 10 (c) 11 (d) 12

344. What is the of centres of circles which touch a given line at a given point ? [CDS (I) 2025 Q44]

(a) line pen*ndicular to thc given line, passing through the given (b) line parallel to the given line

(c) ircle tangent to the given line at the given point (d) losed curve other -than a circle

345. 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (I) 2025 Q45]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

346. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (I) 2025 Q46]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

347. A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (I) 2025 Q47]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x (a) 18 (b)

348. The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (I) 2025 Q48]

(a) Statement- I statement-2 (b) Statement-2 statement-I

(c) Statement-I statement-2 (d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

349. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (I) 2025 Q49]

(a) I only (b) 2 only

(c) oth I and (d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

350. The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz— x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. [CDS (I) 2025 Q50]

(a) 2 only above statements (b) /5 (c) /3 (d) /4

351. A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (I) 2025 Q51]

(a) Their areas are equal (b) The area Of the circle is larger

(c) The area of the square is — times area of circle (d) The area of the square is It times area of circle

352. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (I) 2025 Q52]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

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353. A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (I) 2025 Q53]
(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?
(c) m2 (d) 50

354. In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (I) 2025 Q54]
(a) +1 (b) + (c) l (d) 266+1

355. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (I) 2025 Q55]
(a) — •E (b) litre (c) litre , (d) — litre

356. If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (I) 2025 Q56]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

357. What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (I) 2025 Q57]
(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

358. What is the approximate area of minor segment of the circle ? [CDS (I) 2025 Q58]
(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

359. What is the sum of the radii of the circles ? [CDS (I) 2025 Q59]
(a) 104 cm (b) 2 cm (c) 12 cm (d) 13 cm

360. The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (I) 2025 Q60]
(a) m (b) I cm (c) m (d) m A -

361. A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (I) 2025 Q61]
(a) 121 (b) 120 (c) 119 (d) None of the above

362. In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (I) 2025 Q62]
(a) 20% (b) 40%
(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC=

363. What is the value of LAOC ? [CDS (I) 2025 Q63]
(a) 540 (b) 520 (c) 390 (d) 270

364. The diameter Of a circle circumscribing a Square is 1545 cm. What is the length of the side of the square ? [CDS (I) 2025 Q64]
(a) 360 450 (b) 48
(c) 600 97. 470 670 N In the figure given above, PQ is parallel to RS.
What is Z NMS equal to ? (d) 270

365. Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (I) 2025 Q65]
(a) I Ctn (b) 22 cm
(c) 33 cm (d) 44 W/ 52A 20 58. is equal to — I, then (a)

366. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (I) 2025 Q66]
(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

367. What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (I) 2025 Q67]
(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

368. If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (I) 2025 Q68]
(a) +1 (b) + (c) l (d) 266+1

369. What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (I) 2025 Q69]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

370. What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (I) 2025 Q70]
(a) — square cm (b) 16vTä— — square cm (c) (160 — 8r) square cm (d) — square cm

371. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (I) 2025 Q71]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

372. The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) 2025 Q72]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

373. A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) 2025 Q73]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

374. If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (I) 2025 Q74]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

375. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) 2025 Q75]
(a) 256 (b) 512 (c) 768 (d) 1024

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376. A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (I) 2025 Q76]

(a) 824 (b) 1236 (c) 1628 (d) 2472

377. The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) 2025 Q77]

(a) litre (b) litre , (c) — litre (d) — litre

378. A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) 2025 Q78]

(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

379. A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) 2025 Q79]

(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

380. How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) 2025 Q80]

(a) 420 (b) 210 (c) 200 (d) 168 A -

381. The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (I) 2025 Q81]

(a) 1078 cm (b) 1617 cm (c) 1927 cm (d) 2156 cma

382. A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (I) 2025 Q82]

(a) — cm (b) — cm (c) — cm (d) — cm

383. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) 2025 Q83]

(a) 100 (b) 90 (c) 80 (d) 75

384. Which one of the following statements is correct ? [CDS (I) 2025 Q84]

(a) Mahesh saves more than what Kamal saves (b) Savings of both of them are equal

(c) Kamal saves more than what Mahesh saves (d) It is not possible to determine who saves more

385. The average score of class X is [CDS (I) 2025 Q85]

(a) 150 (b) 300 (c) 450 (d) 600

386. The average score of class Z is [CDS (I) 2025 Q86]

(a) 150 (b) 300 (c) 450 (d) 600

387. What is the initial rate of interest per annum ? [CDS (I) 2025 Q87]

(a) Rs. 1,200 (b) Rs. 1,440 (c) Rs. 1,500 (d) Rs. 1,800

388. If the digits are reversed,' the number is increased by [CDS (I) 2025 Q88]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

389. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (I) 2025 Q89]

(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? (b) 47

(c) 43 (d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

390. ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (I) 2025 Q90]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

391. A line through the vertex A of a parallelogram ABCD meets DC in p and BC produced in Q. If P is the mid-point of DC, then which of the following Ware correct ? [CDS (I) 2025 Q91]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

392. If the diagonals Of a rhombus are X and y, then what is its area ? 2 (b) 4 (c) xy [CDS (I) 2025 Q92]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

393. The area of a rhombus is 336 square cm. Ifthe length ofone of its diagonals is 48 cm, then what is the perimeter of rhombus ? [CDS (I) 2025 Q93]

(a) 200 cm (b) 120 cm (c) IOOcm (d) 90 cm A - 2 sin39— sino

394. ABCD is paralle I square metre square metre square metre square metre a trapezium in which AB is to DC Let E and F be the midpoints on AD and BC respectively. If EF IO cm and AB— DC 4 cm, then what is the value of ABX DC? [CDS (I) 2025 Q94]

(a) 84 square cm (b) 96 square cm (c) 100 square cm (d) 108 square cm

395. A two-digit number is such that the product of the digits is [CDS (I) 2025 Q95]

(a) 1 only (b) 2 only

(c) oth 1 and (d) Neither 1 nor 2 only when Ware correct ?

396. If 63 is added to this number, the digits interchange their places. What is the sum of the digits in the number ? [CDS (I) 2025 Q96]

(a) 1 only (b) 2 only

(c) oth 1 and (d) Neither 1 nor 2 only when Ware correct ?

— SPACE FOR ROUGH WORK —

397. If a = xyp—l, b c aq ¯ ¯ P cP — q is equal to = zxr —1, then [CDS (I) 2025 Q97]
(a) (b) c equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to
(c) c (d) xyz

398. If and b. c 2a : 3b : 7c is equal to [CDS (I) 2025 Q98]
(a) 42 : 105 : 320 (b) 15 : 21 :
(c) 6: 15:40 (d) 30; 21 : 350• 10 = 7: 8, then

399. Two sides Of a parallelogram are IO cm and IS cm. If the altitude corresponding to the side Of length 15 cm is 5 crn, then what is the altitude to the side Of length [CDS (I) 2025 Q99]
(a) 1/64 What is the greatest number which divides 392, 486 and 627 so as to leave the same remainder in each case ? (b) 47
(c) 43 (d) 34 A man walking at the rate 3 km.,'hr crosses a field diagorully in I minute. What is the area of the field ?

400. ABCD is a parallelogram where AC and BD are the diagonals. If ZBAD 600, ZADB 900, then what is BD2 equal to ? [CDS (I) 2025 Q100]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

SECTION : CDS (II) 2024 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

401. HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (II) 2024 Q1]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

402. A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (II) 2024 Q2]
(a) 54 (b) 81
(c) 108 (d) 135 A hollow spherical shell is made of a metal of density 7 g/cm

403. If its internal and extemal radii are 3 cm and 6 cm respectively, then what is the mass of the shell ? (take [CDS (II) 2024 Q3]
(a) 2772 g (b) 3322 g (c) 4433 g (d) 5544 g

404. What is AB2 + 4BQ2 equal to ? [CDS (II) 2024 Q4]
(a) 41•25cm2 (b) 42•25 cm2 (c) 75 cm2 (d) 44•25 cm2

405. What is AQ2 + CP2 equal to ? (b) (c) (d) 1-2 AC2 1-25 AC2 1-5 AC2 A - [CDS (II) 2024 Q5]
(a) 10•05 cm2 (b) 10•15cm2
(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

406. A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (II) 2024 Q6]
(a) 1010 (b) 1015 (c) 1020 (d) 1025

407. ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (II) 2024 Q7]
(a) 443 cm (b) 9 cm (c) I Ocm (d) m

408. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (II) 2024 Q8]
(a) 500 (b) 600 (c) 700 (d) 800

409. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (II) 2024 Q9]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

410. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (II) 2024 Q10]
(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

411. What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (II) 2024 Q11]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

412. The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (II) 2024 Q12]
(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

413. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (II) 2024 Q13]
(a) 120' (b) 135'
(c) 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is (d) following? aeroplane is Observed to be

414. p + q r + s q + r s+p [CDS (II) 2024 Q14]
(a) x3y3 (b) x3y2 (c) Option C (d) Option D

415. 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (II) 2024 Q15]
(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

416. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (II) 2024 Q16]
(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

— SPACE FOR ROUGH WORK —

417. A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (II) 2024 Q17]

(a) The area Of the circle is e qual to that of square

(b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square

(d) No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x (a) 18 (b)

418. The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (II) 2024 Q18]

(a) Statement- I statement-2

(b) Statement-2 statement-I

(c) Statement-I statement-2

(d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

419. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (II) 2024 Q19]

(a) I only

(b) 2 only

(c) oth I and

(d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

420. The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz— x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. [CDS (II) 2024 Q20]

(a) 2 only above statements (b) /5 (c) /3 (d) /4

421. A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (II) 2024 Q21]

(a) Their areas are equal

(b) The area Of the circle is larger

(c) The area of the square is — times area of circle

(d) The area of the square is It times area of circle

422. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (II) 2024 Q22]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

423. x-2+21/2 +23/2 t, x2 20. -4x-10 (p+q)P+q, 512 (d) 12 21. (c—a)(c+a+b) c2 bc—a2 —ab 2(c— a) [CDS (II) 2024 Q23]

(a) 2p

(b) Option B

(c) 1/2 22. ab - ba t, /4SA 24. -Q

(d) 12 21.

424. In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (II) 2024 Q24]

(a) +1 (b) + (c) l (d) 266+1

425. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (II) 2024 Q25]

(a) — •E (b) litre (c) litre , (d) — litre

426. If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (II) 2024 Q26]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

427. What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (II) 2024 Q27]

(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

428. What is the approximate area of minor segment of the circle ? [CDS (II) 2024 Q28]

(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

429. What is the sum of the radii of the circles ? [CDS (II) 2024 Q29]

(a) 104 cm (b) 2 cm (c) 12 cm (d) 13 cm

430. The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (II) 2024 Q30]

(a) m (b) I cm (c) m (d) m A -

431. In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (II) 2024 Q31]

(a) 20%

(b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and

(d) I only Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC=

432. What is the value of LAOC ? [CDS (II) 2024 Q32]

(a) 540 (b) 520 (c) 390 (d) 270

433. The diameter Of a circle circumscribing a Square is 1545 cm. What is the length of the side of the square ? [CDS (II) 2024 Q33]

(a) 360 450

(b) 48

(c) 600 97. 470 670 N In the figure given above, PQ is parallel to RS. What is Z NMS equal to ?

(d) 270

434. Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (II) 2024 Q34]

(a) I Ctn

(b) 22 cm

(c) 33 cm

(d) 44 W/ 52A 20 58. is equal to — I, then (a)

435. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (II) 2024 Q35]

(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

— SPACE FOR ROUGH WORK —

436. What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (II) 2024 Q36]
(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

437. If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (II) 2024 Q37]
(a) +1 (b) + (c) l (d) 266+1

438. What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (II) 2024 Q38]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

439. What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (II) 2024 Q39]
(a) — square cm (b) 16vTä— — square cm (c) (160 — 8r) square cm (d) — square cm

440. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (II) 2024 Q40]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

441. The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (II) 2024 Q41]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

442. A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (II) 2024 Q42]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

443. If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (II) 2024 Q43]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

444. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (II) 2024 Q44]
(a) 256 (b) 512 (c) 768 (d) 1024

445. A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (II) 2024 Q45]
(a) 824 (b) 1236 (c) 1628 (d) 2472

446. The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (II) 2024 Q46]
(a) litre (b) litre , (c) — litre (d) — litre

447. A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (II) 2024 Q47]
(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

448. A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (II) 2024 Q48]
(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

449. How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (II) 2024 Q49]
(a) 420 (b) 210 (c) 200 (d) 168 A -

450. The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (II) 2024 Q50]
(a) 1078 cm (b) 1617 cm (c) 1927 cm (d) 2156 cma

451. A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (II) 2024 Q51]
(a) — cm (b) — cm (c) — cm (d) — cm

452. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (II) 2024 Q52]
(a) 100 (b) 90 (c) 80 (d) 75

453. What is the number of students in the class ? [CDS (II) 2024 Q53]
(a) 100 (b) 105 (c) 110 (d) 120

454. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (II) 2024 Q54]
(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

455. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (II) 2024 Q55]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

456. A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (II) 2024 Q56]
(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

457. Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (II) 2024 Q57]
(a) 4275 (b) 4350 (c) 4375 (d) 4450

— SPACE FOR ROUGH WORK —

458. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (II) 2024 Q58]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

459. Which of the following is/are correctly matched ? [CDS (II) 2024 Q59]

(a) Mean Median (b) Mean Mode

(c) Mean > Median (d) Mean > Mode What is the harmonic mean 25, 40 and 50 ?

460. The Of two number' is 6 10 12 15 Of N:uation• y O a unique x unique X infinite solutions -2+4 Za-l 2.3—28-1 2.8—'28 + 292 •a + — Qy) only O.OETA [CDS (II) 2024 Q60]

(a) k = (b) k = (c) k = (d) k = ±4

461. 3bc c c Sbe + b [d) 3be 4 b3 + then what is x What ig the value Of O ne Ot factors tf — 5x2 — 81 ia ta) equgi to ? a unique x a unique solution x — infinite solutions 2.3-28-1 2.8-'28 291 + [CDS (II) 2024 Q61]

(a) x = 1, y = (b) x = 2, y = (c) x = 3, y = (d) x = 1, y =

462. A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (II) 2024 Q62]

(a) 60 (b) 70 (c) 80 (d) 90

463. Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (II) 2024 Q63]

(a) 264 (b) 201 (c) 132 (d) 105

464. Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (II) 2024 Q64]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

465. HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (II) 2024 Q65]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

466. If A+B= equal to ? x2 —4x+4 2x2 -7X+3 2x-1 2x2+7x-3 2x-1 then what is B 7 the [CDS (II) 2024 Q66]

(a) 30 (b) 45 (c) 60 (d) 75

467. If then what is (a + b) : (a — b) equal to ? [CDS (II) 2024 Q67]

(a) 3 only (b) 5 only (c) —3 or (d) —5 or

468. Consider the following inequalities : a2—b2 a—b > — where a > b > O a2+b2 a a3+b3 a2+b2 only when 2. a2+b2 a+b Which of the above is/are correct ? [CDS (II) 2024 Q68]

(a) 1 only (b) 2 only (c) oth 1 and (d) Neither 1 nor

469. A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (II) 2024 Q69]

(a) 54 (b) 81 (c) 108 (d) 135

470. ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (II) 2024 Q70]

(a) 443 cm (b) 9 cm (c) I Ocm (d) m

471. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (II) 2024 Q71]

(a) 500 (b) 600 (c) 700 (d) 800

472. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (II) 2024 Q72]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

473. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (II) 2024 Q73]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

474. What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (II) 2024 Q74]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

475. The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (II) 2024 Q75]

(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

476. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (II) 2024 Q76]

(a) 120' (b) 135'

(c) 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is (d) following? aeroplane is Observed to be

477. What is the of centres of circles which touch a given line at a given point ? [CDS (II) 2024 Q77]

(a) line pen*ndicular to thc given line, passing through the given (b) line parallel to the given line

(c) ircle tangent to the given line at the given point (d) losed curve other -than a circle

478. 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (II) 2024 Q78]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

— SPACE FOR ROUGH WORK —

479. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (II) 2024 Q79]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

480. A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (II) 2024 Q80]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x (a) 18 (b)

481. The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (II) 2024 Q81]

(a) Statement- I statement-2 (b) Statement-2 statement-I

(c) Statement-I statement-2 (d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

482. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (II) 2024 Q82]

(a) I only (b) 2 only

(c) oth I and (d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

483. A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (II) 2024 Q83]

(a) Their areas are equal (b) The area Of the circle is larger

(c) The area of the square is — times area of circle (d) The area of the square is It times area of circle

484. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (II) 2024 Q84]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

485. A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (II) 2024 Q85]

(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) m2 (d) 50

486. In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (II) 2024 Q86]

(a) +1 (b) + (c) l (d) 266+1

487. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (II) 2024 Q87]

(a) — •E (b) litre (c) litre , (d) — litre

488. If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (II) 2024 Q88]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

489. What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (II) 2024 Q89]

(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

490. What is the approximate area of minor segment of the circle ? [CDS (II) 2024 Q90]

(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

491. What is the sum of the radii of the circles ? [CDS (II) 2024 Q91]

(a) 104 cm (b) 2 cm (c) 12 cm (d) 13 cm

492. The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (II) 2024 Q92]

(a) m (b) I cm (c) m (d) m A -

493. A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (II) 2024 Q93]

(a) 121 (b) 120 (c) 119 (d) None of the above

494. In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (II) 2024 Q94]

(a) 20% (b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only

495. The diameter Of a circle circumscribing a Square is 1545 cm. What is the length of the side of the square ? [CDS (II) 2024 Q95]

(a) 360 450 (b) 48

(c) 600 97. 470 670 N In the figure given above, PQ is parallel to RS.
What is Z NMS equal to ? (d) 270

496. Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (II) 2024 Q96]

(a) I Ctn (b) 22 cm

(c) 33 cm (d) 44 W/ 52A 20 58. is equal to — I, then (a)

497. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (II) 2024 Q97]

(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

498. What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (II) 2024 Q98]

(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

— SPACE FOR ROUGH WORK —

499. If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (II) 2024 Q99]
(a) +1 (b) + (c) l (d) 266+1

500. What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (II) 2024 Q100]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

SECTION : CDS (I) 2024 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

501. I 111. [CDS (I) 2024 Q1]
(a) Option A (b) Option B (c) Option C (d) Nil, 482 5.

502. Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (I) 2024 Q2]
(a) I Ctn (b) 22 cm
(c) 33 cm (d) 44 W/ 52A 20 58. is equal to — I, then (a)

503. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (I) 2024 Q3]
(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

504. What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (I) 2024 Q4]
(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

505. If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (I) 2024 Q5]
(a) +1 (b) + (c) l (d) 266+1

506. What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (I) 2024 Q6]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

507. 10. 11. 1. 12. 13. (b) (c) [CDS (I) 2024 Q7]
(a) Option A (b) Option B (c) Option C (d) Option D

508. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(l cm3 of the metal weighs 9 g) ? [CDS (I) 2024 Q8]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

509. The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) 2024 Q9]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

510. A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) 2024 Q10]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

511. If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (I) 2024 Q11]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

512. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) 2024 Q12]
(a) 256 (b) 512 (c) 768 (d) 1024

513. A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (I) 2024 Q13]
(a) 824 (b) 1236 (c) 1628 (d) 2472

514. The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) 2024 Q14]
(a) litre (b) litre , (c) — litre (d) — litre

515. A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) 2024 Q15]
(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

516. A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) 2024 Q16]
(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

517. How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) 2024 Q17]
(a) 420 (b) 210 (c) 200 (d) 168 A -

518. The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (I) 2024 Q18]
(a) 1078 cm (b) 1617 cm (c) 1927 cm (d) 2156 cma

519. A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (I) 2024 Q19]
(a) — cm (b) — cm (c) — cm (d) — cm

520. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) 2024 Q20]
(a) 100 (b) 90 (c) 80 (d) 75

521. 24. [CDS (I) 2024 Q21]
(a) Option A (b) Option B (c) Option C (d) Option D

— SPACE FOR ROUGH WORK —

522. What is the square root of 23 —4•v'fiS ? (b) 7-3-6 [CDS (I) 2024 Q22]

(a) 29, 17, (b) 31, 17, (c) 19, 37, (d) 23, 31,

523. What is (d) [CDS (I) 2024 Q23]

(a) 30 (b) 45 (c) 60 (d) 75 equal to ?

524. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (I) 2024 Q24]

(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

525. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2024 Q25]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

526. A sum of Rs. 10,000 is deposited for 1 year at the rate of interest 10% com- pounded half yearly. What will be the interest at the end of one year ? [CDS (I) 2024 Q26]

(a) Rs. 1000 (b) RK1025

(c) Rs. 1050 (d) RS. 1100 a mixture of milk and water of

527. A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (I) 2024 Q27]

(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

528. Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (I) 2024 Q28]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

529. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) 2024 Q29]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

530. A tin of oil was — full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was — fill]. Oil of. how many EX)ttles can the tin contain ? (All bottles are of equal volurne) [CDS (I) 2024 Q30]

(a) 35 (b) 40 (c) 45 (d) 50

531. Consider the following in respect of the equation y = l. if x > 1 2. if x < 1 3. y exists for all values of x Which of the above statements is/are correct ? [CDS (I) 2024 Q31]

(a) I only (b) 2 only

(c) I and 2 only l, 2 and Y is/are 14. If (d) 2 minutes

532. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ? (b) 5<k$11 [CDS (I) 2024 Q32]

(a) 2% (b) 8% 20. If k=x—y+2z where —2<-r$l, —l y s 2 and 3 S z «•then which one of the following is correct ?

(c) 5<k$11 21. The number of pairs (d) Zero

533. A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (I) 2024 Q33]

(a) 60 (b) 70 (c) 80 (d) 90

534. Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (I) 2024 Q34]

(a) 264 (b) 201 (c) 132 (d) 105

535. Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (I) 2024 Q35]

(a) Rs. (b) Rs.

(c) Rs. (d) Rs. 14 43. 44. 45. (12-A) AB and CD are two diameters Of a circle of radius r and they are mutually perpendicular. What is the ratio of the area Of the circle to the area of the triangle ACD.? 46. 47. 48.

536. How many circular plates Of diameter d can be taken out of a square plate of side 2d with minimum loss of material ? [CDS (I) 2024 Q36]

(a) Option A (b) Option B (c) Option C (d) Option D

537. HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (I) 2024 Q37]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

538. A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (I) 2024 Q38]

(a) 54 (b) 81 (c) 108 (d) 135

539. How much less amount is allocated to A and C together as compared to D and E together? [CDS (I) 2024 Q39]

(a) 0004 lakhs (b) 0005 lakhs (c) 6008 lakhs (d) 0010 lakhs 45K

540. A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (I) 2024 Q40]

(a) 1010 (b) 1015 (c) 1020 (d) 1025

541. ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (I) 2024 Q41]

(a) 443 cm (b) 9 cm (c) I Ocm (d) m

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542. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (I) 2024 Q42]

(a) 500 (b) 600 (c) 700 (d) 800

543. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2024 Q43]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

544. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (I) 2024 Q44]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

545. What the maximum number of circum-circles that a triangle can have ? (d) Infinite [CDS (I) 2024 Q45]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

546. The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (I) 2024 Q46]

(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

547. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (I) 2024 Q47]

(a) 120' (b) 135'

(c) 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is (d) following? aeroplane is Observed to be

548. What is the of centres of circles which touch a given line at a given point ? [CDS (I) 2024 Q48]

(a) line pen*ndicular to thc given line, passing through the given (b) line parallel to the given line

(c) ircle tangent to the given line at the given point (d) losed curve other -than a circle

549. 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (I) 2024 Q49]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

550. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (I) 2024 Q50]

(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

551. A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (I) 2024 Q51]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x (a) 18 (b)

552. The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (I) 2024 Q52]

(a) Statement- I statement-2 (b) Statement-2 statement-I

(c) Statement-I statement-2 (d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

553. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (I) 2024 Q53]

(a) I only (b) 2 only

(c) oth I and (d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

554. The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz— x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. [CDS (I) 2024 Q54]

(a) 2 only above statements (b) /5 (c) /3 (d) /4

555. A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (I) 2024 Q55]

(a) Their areas are equal (b) The area Of the circle is larger

(c) The area of the square is — times area of circle (d) The area of the square is It times area of circle

556. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (I) 2024 Q56]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

557. A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (I) 2024 Q57]

(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) m2 (d) 50

558. In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (I) 2024 Q58]

(a) +1 (b) + (c) l (d) 266+1

559. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (I) 2024 Q59]

(a) — •E (b) litre (c) litre , (d) — litre

560. If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (I) 2024 Q60]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

561. What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (I) 2024 Q61]

(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

— SPACE FOR ROUGH WORK —

562. What is the approximate area of minor segment of the circle ? [CDS (I) 2024 Q62]
(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

563. The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (I) 2024 Q63]
(a) m (b) I cm (c) m (d) m A -

564. A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (I) 2024 Q64]
(a) 121 (b) 120 (c) 119 (d) None of the above

565. In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (I) 2024 Q65]
(a) 20% (b) 40%
(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC=

566. What is the value of LAOC ? [CDS (I) 2024 Q66]
(a) 540 (b) 520 (c) 390 (d) 270

567. Four equal-sized maximum circular plates are cut off from a square paper sheet Of area 784 square Cm. The circumference of each plate is [CDS (I) 2024 Q67]
(a) I Ctn (b) 22 cm
(c) 33 cm (d) 44 W/ 52A 20 58. is equal to — I, then (a)

568. Four equal discs are placed such that each one touches two others. If the area Of empty space enclosed by them is 150/847 square centimetre, then the radius of each disc is equal to [CDS (I) 2024 Q68]
(a) 7/6 cm (b) 5/6 cm (c) 112 cm (d) 5/11 cm

569. What is area of. largest triangle inscribed in a semi-circle Of radius r units ? [CDS (I) 2024 Q69]
(a) square units (b) 2r2 square units (c) 3? square units (d) 4r2 square units

570. If area of a circle and a square are same, then What is the ratio Of their perimeters ? 2 4 A - [CDS (I) 2024 Q70]
(a) +1 (b) + (c) I (d) 266+1

571. What is the area of the largest square plate cut from a circular disk Of radius one unit ? [CDS (I) 2024 Q71]
(a) 4 square units (b) 245 square units (c) r square units (d) 2 square units

572. What is the least value of n if 194480+nzm4, where m and n are natural numbers? If then What is the value Of ? What is the area Of the region enclosed by three identical circles (each of radius 4 cm) touching each other? [CDS (I) 2024 Q72]
(a) — square cm (b) 16vTä— — square cm (c) (160 — 8r) square cm (d) — square cm

573. A hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 what is the weight of the hemi- sphere(I cm3 of the metal weighs 9 g) ? [CDS (I) 2024 Q73]
(a) 54 g (b) 366 g (c) 122 g (d) 1087t g

574. The material Of a solid cone is converted into the shape of a solid cylinder Of equal radius. If the height of the cylinder is 5 cm. what is the height Of the cone ? [CDS (I) 2024 Q74]
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

575. A Wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used ? [CDS (I) 2024 Q75]
(a) 22 m (b) 20 m (c) 18 m (d) None Of the above

576. If the diameter of a sphere is doubled, how does its surface•area change ? [CDS (I) 2024 Q76]
(a) It increases two times. (b) It increases three times. (c) It increases four times, (d) It increases eight times.

577. What is the number of wax balls, each Of radius I cm, that can be molded out of a sphere of radius 8 cm ? [CDS (I) 2024 Q77]
(a) 256 (b) 512 (c) 768 (d) 1024

578. A drinking glass Of height 24 cm is in the shape of frustum of a cone and diameters of its bottom and top circular ends are 4 cm and 18 cm respectively. If we take capacity Of the glass as xx cm , then what is the value Of x ? [CDS (I) 2024 Q78]
(a) 824 (b) 1236 (c) 1628 (d) 2472

579. The thickness of a cylinder is I foot, the inner radius of the cylinder is 3 feet and height is 7 feet. TO paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder ? [CDS (I) 2024 Q79]
(a) litre (b) litre , (c) — litre (d) — litre

580. A cone of height 16 cm and diameter 14 cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed ? (take [CDS (I) 2024 Q80]
(a) 1540 cm3 (b) 1078cm3 (c) 1048 cm3 (d) 770 cm3

581. A cubical block of side 14cm is sur- mounted by a hemisphere of radius 7 cm. What is the total surface area of the solid thus formed ? (take 9) [CDS (I) 2024 Q81]
(a) 1330 cm2 (b) 1306 cm2 (c) 1296 cm2 (d) 1256 cm2

582. How many silver coins, cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimen- sions 21 cm x 11 cm x 7 cm ? (take 4) [CDS (I) 2024 Q82]
(a) 420 (b) 210 (c) 200 (d) 168 A -

583. The sum of the height and the radius Of a right circular cylinder is 2 1 cm, and the radius is less than the height. If the curved surface area of the cylinder is 616 cm2, then What is the volume Of the cylinder? (Take It 22 / [CDS (I) 2024 Q83]
(a) 1078 cm (b) 1617 cm (c) 1927 cm (d) 2156 cma

— SPACE FOR ROUGH WORK —

584. A sphere of diameter 14 cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 14 cm. If the sphere is completely submerged in water, then how much will the level of water rise ? [CDS (I) 2024 Q84]

(a) — cm (b) — cm (c) — cm (d) — cm

585. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (I) 2024 Q85]

(a) 100 (b) 90 (c) 80 (d) 75

586. The sum of the areas of AAOD and ABOC is equal to the sum of the areas of MOB and ADOC [CDS (I) 2024 Q86]

(a) 924 square units (b) 824 square units

(c) 624 square units (d) 462 square units 99. 6 cm What is the approximate area of the shaded region in the figure given ?

587. For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? [CDS (I) 2024 Q87]

(a) Zem (b) One (c) Two (d) Four

588. If equal to? 195 16 16 then What is If the equation 4x 2 — 2kx + 3k O has equal roots, then What are the values of k? [CDS (I) 2024 Q88]

(a) Zem (b) One (c) Two (d) Four

589. For how many real values of k is 2 + 12kx 16 a "uare for every integer x ? [CDS (I) 2024 Q89]

(a) Zem (b) One (c) Two (d) Four

590. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (I) 2024 Q90]

(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

591. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (I) 2024 Q91]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

592. A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (I) 2024 Q92]

(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

593. Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (I) 2024 Q93]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

594. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (I) 2024 Q94]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

595. The quantity which must be added to (1— x 2) to obtain x 3 is 2X3 + 3X2 + I 2x3+x + x —1 (b) Consider the following statements • [CDS (I) 2024 Q95]

(a) 15 (b) 25 (c) 20 (d) 30 9. 10.

596. How many 200 mm lengths can be cut from IO m of ribbon? [CDS (I) 2024 Q96]

(a) 50 (b) 40 (c) 30 (d) 20

597. What is 262 +972 equal to? @ 272 +932 (b) 342 +932 (C) 822 +412 (d) 792 + 622 [CDS (I) 2024 Q97]

(a) HCF (b) HCF

(c) HCF (d) None Of the above If b is the largest square divisor Of c and a2 divides C, then which one of the following is correct?

598. A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (I) 2024 Q98]

(a) 60 (b) 70 (c) 80 (d) 90

599. Into a conical tent Of radius 8•4m and vertical height m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is [CDS (I) 2024 Q99]

(a) 264 (b) 201 (c) 132 (d) 105

600. Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (I) 2024 Q100]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

SECTION : CDS (II) 2023 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

601. The ratio of two complementary angles is I : [CDS (II) 2023 Q1]

(a) 64 minutes (b) 80 minutes

(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

602. What is the average marks Of all the candidates? [CDS (II) 2023 Q2]

(a) 758 (b) 8 (c) 760. (d) 768'.

603. If two tridngles have equal area, then they are siinilar. Which Of the statements given above is/are corect? [CDS (II) 2023 Q3]

(a) I only (b) 2 only I (c) oth I and (d) Neither I nér

604. AD2 — 3CD2 AC2 = AD2 [CDS (II) 2023 Q4]

(a) 121-7 121-75 (b) 121-75 ABC t, X Y AX.AYur (c) 16 (d) 20 cm

— SPACE FOR ROUGH WORK —

605. What is the sum of their reciprocals ? [CDS (II) 2023 Q5]

(a) 2 (b) 2 (c) 2 (d) 562 + b2

606. 3 2 — + I = k 2 , then which One of the is a possible expression k P — 3X+I [CDS (II) 2023 Q6]

(a) Which Of the above statements is/are correct? (b) I only

(c) 2 only (d) oth I and

607. What is bc(a — b) (a — equal to ? [CDS (II) 2023 Q7]

(a) Which Of the above statements is/are correct? (b) I only

(c) 2 only (d) oth I and

608. Consider the -following data With regard to production Of cars (in lakhs) : Year 2015 Year 2016 Coun Coun Coun Coun [CDS (II) 2023 Q8]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

609. The marks obtained by 5 students are 21, 27, 19, 26, [CDS (II) 2023 Q9]

(a) In Which Of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015? (b) nd E

(c) , C and D only (d) and E

610. Which product is produced least over the years ? [CDS (II) 2023 Q10]

(a) Type (b) Type 111 (c) Type (d) Type V ,'SOA

611. In Which one Of the following pairs of years, the difference in total number Of tablets produced between them is m inimum? [CDS (II) 2023 Q11]

(a) (2003, 2005) (b) (2001, 2005) (c) g003, 2004) (d) (2000, 2002)

612. A vertical pole of length 80m is situated on the horizontal plane. The base of the pole is at P. There are two points A and B such that p, A, B are on the same straight line. Let the angles of elevation of top of the pole from A and B be a and (a>ß) respectively. If PA 64 m and AB 36 m, then what is (a +ß) equal to ? [CDS (II) 2023 Q12]

(a) 600 (b) 900 (c) 1200 (d) 1350 15 A -

613. If a and are complementary angles, then which one Of the following is [CDS (II) 2023 Q13]

(a) mn — I O For the following two (b) 2/3 For the following tvo

(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

614. O S(a-ß) 2' tan(a—ß) (d) 15. sin29cos29 (sec29+cosec20) 16. + 64cos20 = 16 qgi 2' 17. cosece— cote = m sec9— tane = n, cosece+ seco ? [CDS (II) 2023 Q14]

(a) 2h (b) Option B (c) Option C (d) 14

615. Official UPSC CDS Question [CDS (II) 2023 Q15]

(a) 22 m (b) 20 m

(c) 18 m 95. qt«rq (cm2 t ? 16Jä -8T 16Jä - (a) (b) (c) (d) 16 NG 27 - ' g 96. (a) 3-5 km (b) km (c) 5•5 km (d) 6-5 km cot 19 cot 200 cot 700 cot 75' '

616. Staternent—l : I.Rt LMNbea triangle. Let P, Q be the mid- points of the sides LM, LN respectively. If PQ2&MP2+NQ2, then LMN is a fight angled triangle right angled at L. Statement—II : If in a triangle ABC, .4B2 > BC + CA2, then LACB is obtuse. Which of the following is correct in the light of the above statements ? [CDS (II) 2023 Q16]

(a) oth statement — I and statement 11 are correct and statement — 11 is the reason for statement — I (b) oth statement — I and statement — 11 are corect and statemeht — 11 is not the reason for statement — I

(c) Statement—I is true, but state: ment — 11 is false (d) Statement — I is false, but state— ment — 11 is true

617. The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? [CDS (II) 2023 Q17]

(a) ZACB = I right angle (b) LBAC = I right angle (c) LABC I right angle (d) None of ttw abové - A

618. The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? [CDS (II) 2023 Q18]

(a) only (b) 2 only (c) oth 1 and (d) Neither 1 nor 2 - A

619. Consider the follOW'ing statements • If two mangles are equi- angular, then they are similar. [CDS (II) 2023 Q19]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

620. p fi x-2 — sin20x — cos29= o (b) 200 (c) 205 2 3 2. 21. n Mt I (n—l) [CDS (II) 2023 Q20]

(a) 95 (b) nM-k (c) 205 2 3 2. 21. n Mt I (d) Option D

621. (b) 350 (c) 400 (d) 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (II) 2023 Q21]

(a) is always greater than 4 cm but less than 5 cm (b) is always greater than 5 cm

(c) is always less than 4 cm (d) None of the above Consider the following statements

622. E is the midpoint of the median AD of a triangle ABC If BE produced meets the side AC at F, then CF is equal to [CDS (II) 2023 Q22]

(a) /3 (b) 2AC/3 (c) /2 (d) None of the above

623. If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — : If its perimeter is 52 cm, then What is the length Of the smallest side? [CDS (II) 2023 Q23]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

624. How many right angled triangles can formed by joining the vertices of a cuboid ? [CDS (II) 2023 Q24]

(a) 24 (b) 28 (c) 32 (d) None of the above

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625. In a triangle ABC if A is equal to (c) 2 -B- , thenC+2B [CDS (II) 2023 Q25]

(a) e , thenC+2B

(b) 1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only

(d) 2 only

626. ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively such that BX CY = AZ. Which Of the following is/ are correct? I. XYZ is an equilateral triangle. [CDS (II) 2023 Q26]

(a) 24 16 = then the In the figure given above, AD CD What is the value of Z CDB?

(b) 320

(c) 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct?

(d) Q2+AB2 2 pQ2 BP2

627. In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points Of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that Of seventh triangle ? [CDS (II) 2023 Q27]

(a) 256 : I (b) 128 : (c) 64: (d) 16:1

628. The sides of a triangle are given by c + a2 and (b + c) where a +122, a, b, c are positive. What is the area of the triangle equal to ? 2 a2b2 +b2c2 +c2a2 (b) 2 2 3(a2b2 +b2c2 +c2a2) (d) [CDS (II) 2023 Q28]

(a) 367t cm3

(b) 187t cm3

(c) m3 61. The sides of a triangle are given by c + a2 and

(d) 2

629. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ? [CDS (II) 2023 Q29]

(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

630. The difference between their interior angles is [CDS (II) 2023 Q30]

(a)

(b) c equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) c

(d) xyz

631. ABC is a tingle riÖt at B. If ABS 5 cm BC = What is of the p*iølar drawn from B the hypotenuse ? 243 (e) [CDS (II) 2023 Q31]

(a) 32%

(b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?

(c) m2

(d) 50

632. If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? [CDS (II) 2023 Q32]

(a) +1) cm2 (b) + 3) cm2 (c) l) cm2 (d) 266+1) cm2

633. Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side ? (b) 4.3 (c) 16:9 [CDS (II) 2023 Q33]

(a) — •E (b) litre (c) litre , (d) — litre

634. The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (II) 2023 Q34]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

635. ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take [CDS (II) 2023 Q35]

(a) 452 cm2 (b) 440 cm2 (c) 432 cm2 (d) 420 cm2

636. What is the length of the altitude of the triangle drawn from vertex A on BC ? [CDS (II) 2023 Q36]

(a) 10•05 cm2

(b) 10•15cm2

(c) 05 cm2

(d) 11•15cm2 Consider the following for the next three

637. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (II) 2023 Q37]

(a) 12 cm (b) 12•2 cm (c) 12•4 cm (d) 126 cm

638. If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p — q — — r — — p — q) (d) 16 is always [CDS (II) 2023 Q38]

(a) Positive only (b) Negative only (c) Non-positive only (d) Non-negative only

639. The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? [CDS (II) 2023 Q39]

(a) 30 cm (b) 32 cm (c) 36 cm (d) 40 cm

640. The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is [CDS (II) 2023 Q40]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

641. In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? [CDS (II) 2023 Q41]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

642. Let AD be the altitude of a triangle ABC If and (BD — CD) r, then what is BC equal to? (b) (d) qr / p pr / q pq/r [CDS (II) 2023 Q42]

(a) 20%

(b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and

(d) I only

643. ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? [CDS (II) 2023 Q43]

(a) I only (b) 11 only (c) oth I and (d) More information is needed

644. The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? [CDS (II) 2023 Q44]

(a) I only (b) 11 only (c) oth I and (d) Neither I nor

— SPACE FOR ROUGH WORK —

645. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ? [CDS (II) 2023 Q45]

(a) 21 cm (b) 210 cm2 (c) 42 cm (d) 420 cm2

646. ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? (b) p 12 (d) Op/4 [CDS (II) 2023 Q46]

(a) 15 200 73. (b) 10 (c) 11 (d) 12

647. ABC is a triangle right-angled at C. Let p be the midpoint of BC. If AP cm and AB 20 cm, then what is the perimeter of the triangle ABC? [CDS (II) 2023 Q47]

(a) 40 cm (b) 48 cm (c) 50 cm (d) 60 cm

648. A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is [CDS (II) 2023 Q48]

(a) m2 (b) 240 CITI (c) 480 cm2 (d) 6443 cm2

649. DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is [CDS (II) 2023 Q49]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

650. The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is [CDS (II) 2023 Q50]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

651. If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following • statements is/ are correct? part? [CDS (II) 2023 Q51]

(a) 150 (b) 225 (c) 300 (d) 450

652. What is the area of triangle OMN? [CDS (II) 2023 Q52]

(a) 36 square cm (b) 40 square cm (c) 45 square cm (d) 48 square cm

653. In a triangle ABC, 21 cm, BC—20cm and CA = 13 cm. A per- pendicular CD is drawn upon the longest side. What is the area of the triangle BCD ? [CDS (II) 2023 Q53]

(a) 96 square cm (b) 84 square cm (c) 80 square cm (d) 72 square cm

654. Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? [CDS (II) 2023 Q54]

(a) 200 (b) 150

(c) 300 (d) 100 - A 24 (a) (b) (c) (d) (a) 1 (b) 2 (c) 3 (d)

655. HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — — : If its perimeter is 234 52 cm, then What is the length Of the smallest side? [CDS (II) 2023 Q55]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

656. Which of the following is correct in respect Of the equation 3 e)? (Given that is a real number.) u13,— u13, (d) None of the above [CDS (II) 2023 Q56]

(a) e , thenC+2B (b) 1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only (d) 2 only

657. If a and b are negative numbers and c i s a pos itive real num , then Which of the following is/ are c [CDS (II) 2023 Q57]

(a) M < Median (b) GM Median

(c) M = Me dian Suppose for Which one of the corre ct? (d) GM>1/6

658. If is an integer and a, are the roots of 4x2 — such that I < a < 2 4 and 2 < < 3, then how many values can take ? (c) 14 (d) 15 62 +72 +82 +92 +102 [CDS (II) 2023 Q58]

(a) x = 100 (b) x = IO (c) 330 (d) 340

659. Which one Of the following is correct in respect of the number 1729 ? [CDS (II) 2023 Q59]

(a) It cannot be written as the sum Of the cubes of two positive integers (b) It can be written as the Sum Of the cubes of two positive integers in one way only

(c) It can be written as the sum of the cubes Of two positive integers in two ways only (d) It can be wHtten as the sum of the cubes Of two positive integers in three ways Only

660. A hemispherical bowl of internal radius 20 cm contains sauce. The sauce is (o be filled in conical shaped bottles of radius 5 cm and height 8 cm. What is the number of bottles required ? [CDS (II) 2023 Q60]

(a) 100 (b) 90 (c) 80 (d) 75

661. — land 2 1-1? [CDS (II) 2023 Q61]

(a) 34 (b) 90

(c) 127 (d) 129 What is the largest power Of IO that divides the product 29 x 28 x 27 x 2 x I ?

662. A circle touches all the four sides of a quadrilateral ABCD. If AB— 9 cm, BC 8 cm and CD — 12 cm, then what is DA equal to? [CDS (II) 2023 Q62]

(a) 14 cm (b) 13 cm (c) 12 cm (d) I lcm

663. A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself? [CDS (II) 2023 Q63]

(a) IO years (b) 12 years (c) 15 years (d) 20 years

— SPACE FOR ROUGH WORK —

664. What can be said about of 212n—64n, where n integer ? [CDS (II) 2023 Q64]

(a) Last digit is

(b) Last digit is

(c) Last digit is 2 the expansion is a positive 5. 6. 2 Consider the following assumption and two statements : Assumption Statement — I Statement — 11 A number 'ABCDE' is divisible by 11. E-D+C-B+A is divisible by IL

(d) Last two digits are zero

665. A boy is cycling such that the wheels of the Cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is [CDS (II) 2023 Q65]

(a) 5 km/hour (b) 15•84 kn/hour (c) 16 km/hour (d) 36 km/hour

666. Ihe wheels Of a car are of diameter 80 cm each. The car is travelling at a speed of 66 km,'hour. What is the number of complete revolutions each wheel makes in IO minutes ? [CDS (II) 2023 Q66]

(a) 4275 (b) 4350 (c) 4375 (d) 4450

667. A semi circular thin sheet of a of diameter 28 cm is bent and an open conical cup is made. What is the capacity of the cup ? 1000 [CDS (II) 2023 Q67]

(a) m3 (b) 3006 cm3 (c) m3 1078 (d) m3 - A

668. A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height IO cm each that can be packed in the drum ? [CDS (II) 2023 Q68]

(a) 60 (b) 70 (c) 80 (d) 90

669. Rain water from ao roof 22 m x 20 m drains into a cylindrical vessel having diameter of base 2 m and height 3-5 m. If the vessel is just full, what is the rainfall ? [CDS (II) 2023 Q69]

(a) Rs. (b) Rs. (c) Rs. (d) Rs.

670. HOW many cubic metre Of earth is to be dug out to dig a well of radius 14 m and [CDS (II) 2023 Q70]

(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

671. A hemispherical bowl of intemal radius 18 cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3 cm and internal height 4 cm. What is the number of bottles used to empty the bowl ? [CDS (II) 2023 Q71]

(a) 54 (b) 81 (c) 108 (d) 135

672. A 4-digit number N has exactly 15 distinct divisors. What is the total number of distinct divisors of ? [CDS (II) 2023 Q72]

(a) 16 (b) 30 (c) 45 (d) 225

673. A trolley with two wheels one metre apan is moved clockwise on the circular track around a ground with radius 50 m (described by right wheel). If the size of each wheel is of 1 foot radius and the right wheel turns 1000 times, how many times will the other wheel tum ? [CDS (II) 2023 Q73]

(a) 1010 (b) 1015 (c) 1020 (d) 1025

674. What is the locus of the centre of the front wheel of the bicycle whose diameter is d ? The centroid Of a triangle ABC is 8 cm from the vertex A. What is the length Of the median of the triangle through A ? [CDS (II) 2023 Q74]

(a) Option A

(b) Option B

(c) Option C

(d) line parallel to the path of the wheel of the bicycle at a height d cm A line parallel to the path of the wheel of the bicycle at a height d/2 cm A circle of radius d/2 cm A circle of radius d cm 13.

675. ABC is a triangle right-angled at C. A semicircle is drawn on AB as diameter. Let P be any point on AC produced such that AP = AB IO cm. Further, B and P are joined. If BC 8 cm, then what is BP equal to ? [CDS (II) 2023 Q75]

(a) 443 cm (b) 9 cm (c) I Ocm (d) m

676. In the figure given above, O is the centre Of a circle circumscribing a quadrilateral ABCD. If AB BC and ZBAC 400, then what is Z A DC equal to ? [CDS (II) 2023 Q76]

(a) 500 (b) 600 (c) 700 (d) 800

677. ABCD is a quadrilateral, the sides of Which touch a circle. Which one Of the following is correct? [CDS (II) 2023 Q77]

(a) The area Of the circle is e qual to that of square

(b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square

(d) No conclusiion can be drawn 87• If

678. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p, then the area of the polygon is (b) 128 (b) (d) 2 None of the above [CDS (II) 2023 Q78]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

679. The radius of circum-circle of a right- angled triangle is 10cm and the altitude drawn to the hypotenuse is 8 cm. What is the area of the triangle ? [CDS (II) 2023 Q79]

(a) m2 (b) 80 cm2 (c) 100 cm2 (d) 120 cm2

680. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of the (b) [CDS (II) 2023 Q80]

(a) 120'

(b) 135'

(c) 150• 52. If A, C, D are the angles of a cyclic quadrilateral, then what is the value Of SS. What is

(d) following? aeroplane is Observed to be

681. What is the of centres of circles which touch a given line at a given point ? [CDS (II) 2023 Q81]

(a) line pen*ndicular to thc given line, passing through the given

(b) line parallel to the given line

(c) ircle tangent to the given line at the given point

(d) losed curve other -than a circle

682. 5 km/h 365 km/h 38-5 krn/h 40-5 km/h c In the figure given above, the side of square ABCD is 7 cm. What is the area of the shaded portion, formed by the arcs of the circles with centre at C and A ? [CDS (II) 2023 Q82]

(a) 7 crn2 (b) 28 cm2 (c) 14 cm2 (d) 21 Crn2 - A

— SPACE FOR ROUGH WORK —

683. A wire Of length 36 cm is bent in the form of a semicircle. What is the radius Of the semicircle? [CDS (II) 2023 Q83]
(a) 9 Crn (b) 8 Crn (c) 7 Crn (d) 6 Ctn

684. A circle and a souare have the same perimeter. Which one of the following is correct? [CDS (II) 2023 Q84]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If (x 4 + x — 32? What is One of the values of (x (a) 18 (b)

685. The opposite angles of a cyclic quadrilateral are supplementary. Angle subtended by an arc at the centre is double the angle subtended by it at any point On the remaining part Of the circle. Which one of the following is correct in respect of the above statements ? [CDS (II) 2023 Q85]
(a) Statement- I statement-2 (b) Statement-2 statement-I
(c) Statement-I statement-2 (d) Neither statement-I statement-2 nor statement-2 statement-I (a) (c) (d)

686. The four points B, C, E and D may lie on a circle. Which of the above Statements is/are 95. (b) 17/8 correct ? [CDS (II) 2023 Q86]
(a) I only (b) 2 only
(c) oth I and (d) Neither 1 nor 2 and 81 equal to ? (a) (c) (d) 17/16 17/4 15/4 3, then what is x (a)

687. The chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Which one Of the following is correct? y(2z— y) xa y(2Z— y) = 4X2 (b) 2ygz-y1 x2 (c) 4yPz— x2 (d) . In a triangle ABC, ZB-2ZC=2ZA. [CDS (II) 2023 Q87]
(a) 2 only above statements (b) /5 (c) /3 (d) /4

688. A circle and a square have the same perimeter. Which one of the following is correct ? [CDS (II) 2023 Q88]
(a) Their areas are equal (b) The area Of the circle is larger
(c) The area of the square is — times area of circle (d) The area of the square is It times area of circle

689. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O If AB is the diameter and CD is parallel and half of AB and if the clrcle completes one rotation about the centre O, then the locus of the middle point of CD is a circle of radius 3r 2r - A 20 2-vfir 3 2 , then what is [CDS (II) 2023 Q89]
(a) 4275 (b) 4350 (c) 4375 (d) 4450

690. A piu of wim of 33 is into an rc of a ofradius 14 is the angle by am at ca&e of circle ? (c) 1350 (d) 1500 C - SDHY T-MTÄ [CDS (II) 2023 Q90]
(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?
(c) m2 (d) 50

691. In a circle of radius 5 cm, AB and AC are two chords such that AB 8 cm. What is the length of chord BC? [CDS (II) 2023 Q91]
(a) +1 (b) + (c) I (d) 266+1

692. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle ? [CDS (II) 2023 Q92]
(a) — •E (b) litre (c) litre , (d) — litre

693. If an arc of a circle of radius 6 cm subtends a central angle measuring 300, then which one of the following is an approximate length of the arc ? [CDS (II) 2023 Q93]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

694. What is the area of the shaded region in the given figure, ifthe radius of each Of the circles is 2 cm ? [CDS (II) 2023 Q94]
(a) — cm2 (b) — cm2 (c) — E cm2 (d) m2

695. What is the approximate area of minor segment of the circle ? [CDS (II) 2023 Q95]
(a) 2•71 cm2 (b) 2•42 cm2 (c) 1•91 cm2 (d) 1•71cm2 A -

696. What is the sum of the radii of the circles ? [CDS (II) 2023 Q96]
(a) 104 cm (b) 2 cm (c) 12 cm (d) 13 cm

697. The area of the circle circumscribing three identical circles touching each other is square cm. What is the radius of one of the smaller circles ? [CDS (II) 2023 Q97]
(a) m (b) I cm (c) m (d) m A -

698. A circle is inscribed in a triangle ABC right-angled at B. If AB 5 cm and BC = 12 cm, then what is the radius of the circle ? [CDS (II) 2023 Q98]
(a) 121 (b) 120 (c) 119 (d) None of the above

699. In a circle of radius 14 cm, APB is a shorter arc and P is the midpoint of the arc. Ikt C be the midpoint Of the chord AB and PC 7 cm. What is the length Of chord AP ? the (d) 14 cm 1 P.T.O. [CDS (II) 2023 Q99]
(a) 20% (b) 40%
(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only Let O be the centre of a circle. Let A, B and C lie on the circle such that LABC=

700. What is the value of LAOC ? [CDS (II) 2023 Q100]
(a) 540 (b) 520 (c) 390 (d) 270

SECTION : CDS (I) 2023 : ELEMENTARY MATHEMATICS REAL TEST PAPER (100 QUESTIONS)

701. 4. (—a +b2+c 5. 6. 7. 314th 50% [CDS (I) 2023 Q1]
(a) Option A (b) Option B (c) Option C (d) Option D

702. The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (I) 2023 Q2]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

— SPACE FOR ROUGH WORK —

703. ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take [CDS (I) 2023 Q3]

(a) 452 cm2 (b) 440 cm2 (c) 432 cm2 (d) 420 cm2

704. What is the length of the altitude of the triangle drawn from vertex A on BC ? [CDS (I) 2023 Q4]

(a) 10•05 cm2 (b) 10•15cm2

(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

705. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (I) 2023 Q5]

(a) 12 cm (b) 12•2 cm (c) 12•4 cm (d) 126 cm

706. If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p — q — — r — — p — q) (d) 16 is always [CDS (I) 2023 Q6]

(a) Positive only (b) Negative only (c) Non-positive only (d) Non-negative only

707. The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? [CDS (I) 2023 Q7]

(a) 30 cm (b) 32 cm (c) 36 cm (d) 40 cm

708. The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is [CDS (I) 2023 Q8]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

709. In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? [CDS (I) 2023 Q9]

(a) 1100 (b) 1200 (c) 1300 (d) 1400

710. Let AD be the altitude of a triangle ABC If and (BD — CD) r, then what is BC equal to? (b) (d) qr / p pr / q pq/r [CDS (I) 2023 Q10]

(a) 20% (b) 40%

(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only

711. ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? [CDS (I) 2023 Q11]

(a) I only (b) 11 only (c) oth I and (d) More information is needed

712. The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? [CDS (I) 2023 Q12]

(a) I only (b) 11 only (c) oth I and (d) Neither I nor

713. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ? [CDS (I) 2023 Q13]

(a) 21 cm (b) 210 cm2 (c) 42 cm (d) 420 cm2

714. ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? (b) p 12 (d) Op/4 [CDS (I) 2023 Q14]

(a) 15 200 73. (b) 10 (c) 11 (d) 12

715. 110, 41, 43, 95, 127, 99, 61, 92, 71, 93, 110, 36. 93 t, [CDS (I) 2023 Q15]

(a) Option A (b) Option B (c) Option C (d) 18. 19. 20.

716. A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is [CDS (I) 2023 Q16]

(a) m2 (b) 240 CITI (c) 480 cm2 (d) 6443 cm2

717. DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is [CDS (I) 2023 Q17]

(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

718. The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is [CDS (I) 2023 Q18]

(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

719. If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following • statements is/ are correct? part? [CDS (I) 2023 Q19]

(a) 150 (b) 225 (c) 300 (d) 450

720. What is the area of triangle OMN? [CDS (I) 2023 Q20]

(a) 36 square cm (b) 40 square cm (c) 45 square cm (d) 48 square cm

721. 24. , 25. p3+3bp 2a 3a 910@ ? [CDS (I) 2023 Q21]

(a) Option A (b) Option B (c) Option C (d) Option D

722. Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? [CDS (I) 2023 Q22]

(a) 200 (b) 150

(c) 300 (d) 100 - A 24 (a) (b) (c) (d) (a) 1 (b) 2 (c) 3 (d)

723. HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — — : If its perimeter is 234 52 cm, then What is the length Of the smallest side? [CDS (I) 2023 Q23]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

724. Consider the following statements : [CDS (I) 2023 Q24]

(a) (b) c equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to

(c) c (d) xyz

725. What is åe of bikers in mad 2017 ? [CDS (I) 2023 Q25]

(a) 163 (b) 152 (c) 141 (d) 134

— SPACE FOR ROUGH WORK —

726. If of a Üid its rec9mcal is , thn numbers [CDS (I) 2023 Q26]
(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

727. If p divi&s qr if p divides q, p must divi& r. 2' 2' WNch of the [CDS (I) 2023 Q27]
(a) 1 only (b) 96" cm2 (c) 2 only (d) oå 1 2 69T cm2

728. The number 2x3*5x7x11+1 is [CDS (I) 2023 Q28]
(a) prime number (b) not a prime, but power of a prime
(c) not a power of a prime, but a composite even number (d) not a of a prime, but a composite odd number

729. Consider the following statements : I. The line segment AP divides the area of the triangle ABC into two equal parts [CDS (I) 2023 Q29]
(a) I and 11 only (b) 11 and Ill only
(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

730. In which year the difference in their prices exceeded 0000 for the first time ? [CDS (I) 2023 Q30]
(a) 2012 (b) 2013 (c) 2014 (d) 2015

731. If the annual income of X is 20% more than that of then the income of Y is less than that of X by pO/o. What is the value of p ? [CDS (I) 2023 Q31]
(a) 10 (b) 16i (c) 36 (d) 27

732. If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What • the ratio Of their areas ? (c) 81 16 / 26A (d' 27 : [CDS (I) 2023 Q32]
(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square
(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

733. The ratio of two complementary angles is I : [CDS (I) 2023 Q33]
(a) 64 minutes (b) 80 minutes
(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

734. The ratio of two complementary angles is I : [CDS (I) 2023 Q34]
(a) 64 minutes (b) 80 minutes
(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

735. Colour of the eye Of the person [CDS (I) 2023 Q35]
(a) 21 (b) 24
(c) 36 (d) 42 Which one among the following is not correct ?

736. Number of times a }xrson correctly hits a target in ten attempts Which of the following in respect of the above is correct ? [CDS (I) 2023 Q36]
(a) I is a continuous variable, 2 is not a variable and 3 is a discrete variable (b) 1 is a continuous variable, 2 and 3 are discrete variables
(c) I and 3 are discrete variables and 2 is not a variable (d) •l, 2 and 3 all are discrete variables Consider the followiög •statements :

737. The data collected by investigator to be used by himself are called primary data [CDS (I) 2023 Q37]
(a) 21 (b) 24
(c) 36 (d) 42 Which one among the following is not correct ?

738. Consider the following : l. 10+ I [CDS (I) 2023 Q38]
(a) p (b) 3 and —4 (c) —3 and—4 (d) —3 and

739. If a set Acontains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does A u B contain ? [CDS (I) 2023 Q39]
(a) 130 (b) 100 (c) 80 (d) 70

740. If two factors of — 2a3 — 9a2 + 2a + 8 are (a + l) and (a — l), then what are the Other two factors ? [CDS (I) 2023 Q40]
(a) — 2) and (a + 4) (b) + 2) and (a + 4) (c) + 2) and (a — 4) (d) — 2) and (a —4)

741. If 0.764 y 1.236 x, then what is the Y¯x value Of [CDS (I) 2023 Q41]
(a) x- (b) x =
(c) x = 11.5 If 0.764 y = 1.236 x, Y—x value Of (d) 0.764

742. What is the sum of the linear factors (in x and y) of the expression 2x-2 + — 3y2 ? [CDS (I) 2023 Q42]
(a) 2t-3y (b) 3x-2y (c) 3x + 2y (d) 2r+3y

743. For which values of k, does the equation x2 — kx + 2 0 have real and distinct solutions ? (b) only (c) k>2v5 only (d) k<-24i or k>2Vi [CDS (I) 2023 Q43]
(a) only (b) k>2v5 only
(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

744. If a + D + Y aß+ "y + ya, then what is (1 equal to? [CDS (I) 2023 Q44]
(a) only (b) k>2v5 only
(c) k<-24i or k>2Vi 12. If a + D + Y aß+ "y + ya, then what is (d) I—aßy

745. How many zeros are there in the product 150 x 249 x 348* x 501 ? [CDS (I) 2023 Q45]
(a) 262 (b) 261 (c) 246 (d) 235

746. If 96-64a3 3 = what is alt + 4a3 equal to ? then [CDS (I) 2023 Q46]
(a) 30 (b) 45 (c) 60 (d) 75

— SPACE FOR ROUGH WORK —

747. A vertical pole of length 80m is situated on the horizontal plane. The base of the pole is at P. There are two points A and B such that p, A, B are on the same straight line. Let the angles of elevation of top of the pole from A and B be a and (a>ß) respectively. If PA 64 m and AB 36 m, then what is (a +ß) equal to ? [CDS (I) 2023 Q47]

(a) 600 (b) 900 (c) 1200 (d) 1350 15 A -

748. If a and are complementary angles, then which one Of the following is [CDS (I) 2023 Q48]

(a) mn — I O For the following two (b) 2/3 For the following tvo

(c) 1 P.T.O. For the following two (d) 2/3 sina 4 u sin # cosa cosß

749. m, then the height Of the tower is close to [CDS (I) 2023 Q49]

(a) 86.6m (b) 84.6 m

(c) 82.6 m (d) 80.6 m 16 In a triangle ABC, LBCA = 600 and AB2 = BC2 + CA2 + X. What is X equal (a) (b) (c) (d) Zero

750. Staternent—l : I.Rt LMNbea triangle. Let P, Q be the mid- points of the sides LM, LN respectively. If PQ2&MP2+NQ2, then LMN is a fight angled triangle right angled at L. Statement—II : If in a triangle ABC, .4B2 > BC + CA2, then LACB is obtuse. Which of the following is correct in the light of the above statements ? [CDS (I) 2023 Q50]

(a) oth statement — I and statement 11 are correct and statement — 11 is the reason for statement — I (b) oth statement — I and statement — 11 are corect and statemeht — 11 is not the reason for statement — I

(c) Statement—I is true, but state: ment — 11 is false (d) Statement — I is false, but state— ment — 11 is true

751. The median BD of ghe triangle ABC meets at D. If then which one Of the following is correct ? [CDS (I) 2023 Q51]

(a) ZACB = I right angle (b) LBAC = I right angle (c) LABC I right angle (d) None of ttw abové - A

752. The perimeter of a triangle is greater than the sum of the lengths of its three medians. Which of the statemepts given above is/are correct ? [CDS (I) 2023 Q52]

(a) only (b) 2 only (c) oth 1 and (d) Neither 1 nor 2 - A

753. Consider the follOW'ing statements • If two mangles are equi- angular, then they are similar. [CDS (I) 2023 Q53]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

754. Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (I) 2023 Q54]

(a) is always greater than 4 cm but less than 5 cm (b) is always greater than 5 cm

(c) is always less than 4 cm (d) None of the above Consider the following statements

755. (b) 350 (c) 400 (d) 45 Let ABC be a triangle with AB = 3 Cm, AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct ? [CDS (I) 2023 Q55]

(a) is always greater than 4 cm but less than 5 cm (b) is always greater than 5 cm

(c) is always less than 4 cm (d) None of the above Consider the following statements

756. E is the midpoint of the median AD of a triangle ABC If BE produced meets the side AC at F, then CF is equal to [CDS (I) 2023 Q56]

(a) /3 (b) 2AC/3 (c) /2 (d) None of the above

757. If every side Of an equilateral triangle is doubled, then the area Of new triangle becomes k times the area Of the Old One. What is k equal to? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — : If its perimeter is 52 cm, then What is the length Of the smallest side? [CDS (I) 2023 Q57]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

758. How many right angled triangles can formed by joining the vertices of a cuboid ? [CDS (I) 2023 Q58]

(a) 24 (b) 28 (c) 32 (d) None of the above

759. In a triangle ABC if A is equal to (c) 2 -B- , thenC+2B [CDS (I) 2023 Q59]

(a) e , thenC+2B (b) 1 0+1 cos26 — 3cos8+2 sin2 68. Which of the following is correct in respect Of the equation 3 e

(c) only (d) 2 only

760. ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively such that BX CY = AZ. Which Of the following is/ are correct? I. XYZ is an equilateral triangle. [CDS (I) 2023 Q60]

(a) 24 16 = then the In the figure given above, AD CD What is the value of Z CDB? (b) 320

(c) 640 BC ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct? (d) Q2+AB2 2 pQ2 BP2

761. In an equilateral triangle another equilateral triangle is drawn inside joining the mid-points Of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that Of seventh triangle ? [CDS (I) 2023 Q61]

(a) 256 : I (b) 128 : (c) 64: (d) 16:1

762. The sides of a triangle are given by c + a2 and (b + c) where a +122, a, b, c are positive. What is the area of the triangle equal to ? 2 a2b2 +b2c2 +c2a2 (b) 2 2 3(a2b2 +b2c2 +c2a2) (d) [CDS (I) 2023 Q62]

(a) 367t cm3 (b) 187t cm3

(c) m3 61. The sides of a triangle are given by c + a2 and (d) 2

763. What is the area of a triangle with sides Of length 12 cm, 13 cm and 5 cm ? [CDS (I) 2023 Q63]

(a) 30 crn2 (b) 35 cm2 (c) 40 cm2 (d) 42

— SPACE FOR ROUGH WORK —

764. The difference between their interior angles is [CDS (I) 2023 Q64]
(a) (b) c equation 19. value of 20. zxr — 1, then aq ¯ ¯ P cP — q is equal to
(c) c (d) xyz

765. ABC is a tingle riÖt at B. If ABS 5 cm BC = What is of the p*iølar drawn from B the hypotenuse ? 243 (e) [CDS (I) 2023 Q65]
(a) 32% (b) 30% Sl. If the Of h+ue of a trian* is, IO is tie of awh a riørt mgld ?
(c) m2 (d) 50

766. If the angles of a triangle are 300 and 450 and the included side is + 1) cm, then what is the area of the triangle ? [CDS (I) 2023 Q66]
(a) +1) cm2 (b) + 3) cm2 (c) l) cm2 (d) 266+1) cm2

767. Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side ? (b) 4.3 (c) 16:9 [CDS (I) 2023 Q67]
(a) — •E (b) litre (c) litre , (d) — litre

768. The lengths of sides of a triangle are where If one of the angles is 900, then what are the minimum integral values of x, y, z respectively ? [CDS (I) 2023 Q68]
(a) — AB (b) B2 (c) 1, 2,3 82. 83. (d) 4

769. ABC is a triangle right-angled at B with AB 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surmce area of the cone so formed ? (take [CDS (I) 2023 Q69]
(a) 452 cm2 (b) 440 cm2 (c) 432 cm2 (d) 420 cm2

770. What is the length of the altitude of the triangle drawn from vertex A on BC ? [CDS (I) 2023 Q70]
(a) 10•05 cm2 (b) 10•15cm2
(c) 05 cm2 (d) 11•15cm2 Consider the following for the next three

771. In , a right triangle ABC, BD is perpendicular on hypotenuse AC If AC 9 cm and AD 4 cm, then what is AB+ BC approximately equal to? [CDS (I) 2023 Q71]
(a) 12 cm (b) 12•2 cm (c) 12•4 cm (d) 126 cm

772. If p, q and r are the lengths (in cm) of the sides of a right-angled triangle, then (p — q — — r — — p — q) (d) 16 is always [CDS (I) 2023 Q72]
(a) Positive only (b) Negative only (c) Non-positive only (d) Non-negative only

773. The base of a right-angled triangle is times the height of triangle. If the area of the triangle is 54 square cm, then what is the perimeter of the triangle ? [CDS (I) 2023 Q73]
(a) 30 cm (b) 32 cm (c) 36 cm (d) 40 cm

774. The ratio of sum of interior angles to sum of exterior angles of a regular polygon of n sides is [CDS (I) 2023 Q74]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

775. In a right triangle ABC, 900 and AD is perpendicular to BC. If ECAD 600 and BC = 6 cm, then what is AB equal to ? [CDS (I) 2023 Q75]
(a) 1100 (b) 1200 (c) 1300 (d) 1400

776. Let AD be the altitude of a triangle ABC If and (BD — CD) r, then what is BC equal to? (b) (d) qr / p pr / q pq/r [CDS (I) 2023 Q76]
(a) 20% (b) 40%
(c) None of the above 61. Let AD be the altitude of a triangle ABC If and (d) I only

777. ABC is a triangle right angled at B. P is the midpoint Of AB and Q is the mid- point of BC. Consider the following : I. AQ units 11. units Which of the atxwe is/ are required to determine the area Of the triangle? [CDS (I) 2023 Q77]
(a) I only (b) 11 only (c) oth I and (d) More information is needed

778. The quadrilateral AAY is a The area Of the quadrilateral AZXY is half Of the area Of the triangle Which Of the statements given above is/ are correct? [CDS (I) 2023 Q78]
(a) I only (b) 11 only (c) oth I and (d) Neither I nor

779. In a triangle ABG ZA 300, AB 7 cm and AC• 12 Ctn. What is the area of the triangle ABC ? [CDS (I) 2023 Q79]
(a) 21 cm (b) 210 cm2 (c) 42 cm (d) 420 cm2

780. ABC is a triangle right angled at B. D is a point on AC such that BD is perpendicular to AC If AB p and BC v'äp, then what is BD equal to? (b) p 12 (d) Op/4 [CDS (I) 2023 Q80]
(a) 15 200 73. (b) 10
(c) 11 (d) 12 The difference between an interior angle and an exterior angle of a regular polygon is

781. What is the number of sides of the polygon? 9 (b) 10 (c) 11 (d) 12 ,'45A [CDS (I) 2023 Q81]
(a) 15 200 73. (b) 10 (c) 11 (d) 12

782. ABC is a triangle right-angled at C. Let p be the midpoint of BC. If AP cm and AB 20 cm, then what is the perimeter of the triangle ABC? [CDS (I) 2023 Q82]
(a) 40 cm (b) 48 cm (c) 50 cm (d) 60 cm

783. A square and an equilateral triangle have equal perimeter. If the diagonal of .the square is 12-45 cm, then the area of the triangle is [CDS (I) 2023 Q83]
(a) m2 (b) 240 CITI (c) 480 cm2 (d) 6443 cm2

784. DP and when produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DC? is [CDS (I) 2023 Q84]
(a) 20 square t (b) 30 square unit (c) 40 square unit (d) None of the above

785. The base Of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then the area Of the triangle is [CDS (I) 2023 Q85]
(a) 9600 square units (b) 10000 square units (c) 12000 square units (d) None of the above

— SPACE FOR ROUGH WORK —

786. If the perimeter of an isosceles right triangle is 4 (2 cm, then What is its area in square cm? The 8 16 24 diagonal of a square is 120 cm and the area of an equilateral triangle is 64" Square Cm. Which Of the following • statements is/ are correct? part? [CDS (I) 2023 Q86]

(a) 150 (b) 225 (c) 300 (d) 450

787. What is the area of triangle OMN? [CDS (I) 2023 Q87]

(a) 36 square cm (b) 40 square cm (c) 45 square cm (d) 48 square cm

788. In a triangle ABC, 21 cm, BC—20cm and CA = 13 cm. A per- pendicular CD is drawn upon the longest side. What is the area of the triangle BCD ? [CDS (I) 2023 Q88]

(a) 96 square cm (b) 84 square cm (c) 80 square cm (d) 72 square cm

789. Frequency polygon Frequency curve Frequency distribution All of the above D E Three straight lines X, Y and Z are parallel and the angles are as shown in the figure above. What is ZAFB equal to ? [CDS (I) 2023 Q89]

(a) 200 (b) 150

(c) 300 (d) 100 - A 24 (a) (b) (c) (d) (a) 1 (b) 2 (c) 3 (d)

790. HOW many pairs Of positive integers m and n satisw the equation m n 12 where n is an odd integer less than 60? 17 (d) . The 7 4 3 sides Of a triangle are in the ratio — — : If its perimeter is 234 52 cm, then What is the length Of the smallest side? [CDS (I) 2023 Q90]

(a) 9 cm (b) I O Crn (c) 11 cm (d) 12 cm

791. Which one of the following statements is correct ? [CDS (I) 2023 Q91]

(a) Mahesh saves more than what Kamal saves (b) Savings of both of them are equal

(c) Kamal saves more than what Mahesh saves (d) It is not possible to determine who saves more

792. The average score of class X is [CDS (I) 2023 Q92]

(a) 150 (b) 300 (c) 450 (d) 600

793. The average score of class Z is [CDS (I) 2023 Q93]

(a) 150 (b) 300 (c) 450 (d) 600

794. If the average marks Of the complete class is 72, then what is the average marks Of the girls ? [CDS (I) 2023 Q94]

(a) 64 (b) 65 (c) 68 (d) 74

795. Consider the following statements : I. The line segment AP divides the area of the triangle ABC into two equal parts [CDS (I) 2023 Q95]

(a) I and 11 only (b) 11 and Ill only

(c) I and Ill only (d) 1, 11 and 111 Consider the following for the next two

796. A's salary was increased by and then decreased by 20%. On the whole A's salary is increased by [CDS (I) 2023 Q96]

(a) 100066-1 (b) 10000m (c) 100+-6 (d) 24.16 m

797. In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. nere were Only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is [CDS (I) 2023 Q97]

(a) 3600 (b) 6200 (c) 6028 (d) 6400

798. Le g nd b be real + = 32 50 + b) 7 (d) be 2 one Of the 61%wing ? [CDS (I) 2023 Q98]

(a) e 2 one Of the 61%wing ? 13. HCF numbes is 12. Wych me Of xver LCM ? (b) 80

(c) 60 36 • (d) 24

799. If two corresponding sides Of two similar triangles, are in the ratio 9 : 4, then What • the ratio Of their areas ? (c) 81 16 / 26A (d' 27 : [CDS (I) 2023 Q99]

(a) The area Of the circle is e qual to that of square (b) The area o' the circle is larger than that Of square

(c) The area of the circle is less than that bf square (d) No conclusiion can be drawn 87• If

800. The ratio of two complementary angles is I : [CDS (I) 2023 Q100]

(a) 64 minutes (b) 80 minutes

(c) 96 minutes (d) 120 minutes The pair of numbers which are relatively prime to each other is :

— SPACE FOR ROUGH WORK —